Forcing the Π^1_n-Uniformization Property

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Abstract

We generically construct a model in which the Π^1_3-uniformization property is true, thus lowering the best known consistency strength from the existence of $M_1^{\#}$ to just ZFC. The forcing construction also works over canonical inner models with Woodin cardinals, which yields, for the first time, universes where the Π^1_{2n}-uniformization property holds, thus producing models which contradict the natural PD-induced pattern.

1 Introduction

The question of finding nicely definable choice functions for a definable family of sets is an old and well-studied subject in descriptive set theory; see, for instance, [12, 17]. The uniformization problem, first mentioned by N. Lusin in 1930 (see [14]), asks to find choice functions which lie at the same projective level as the set they aim to uniformize. Recall that for $A \subseteq 2^\omega \times 2^\omega$, a *uniformization* of A is a function

$$f : \mathrm{pr}_1(A) \longrightarrow 2^\omega$$

such that

$$\mathrm{graph}(f) \subseteq A.$$

Equivalently, for every $x \in \mathrm{pr}_1(A)$ one has $(x, f(x)) \in A$.

Definition 1.1. *We say that a pointclass Γ has the uniformization property iff every element of Γ admits a uniformization in Γ.*

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It is a classical result due to M. Kondo [13] that lightface Π^1_1-sets do have the uniformization property, this also yields the uniformization property for Σ^1_2-sets. This is as much as ZFC can prove about uniformization. In the constructible universe L, as shown by J. Addison in [1], Σ^1_n does have the uniformization property for $n \geq 3$, which follows from the existence of a Σ^1_2-good wellorder of the reals; for background on projective wellorders see also [2, 4]. Thus the Π^1_n-uniformization fails for $n \geq 3$. On the other hand, by the celebrated results of Y. Moschovakis (see [16], Theorem 1), $\mathbf{\Delta^1_{2n}}$-projective determinacy implies Π^1_{2n+1}-uniformization, yet the determinacy assumption exceeds in logical strength ZFC. It is known due to H. W. Woodin (see [18],), that $\mathbf{\Delta^1_2}$-projective determinacy implies the existence of $M_1^{\#}$, hence yields an inner model with a Woodin cardinal. As with other regularity properties of the reals like Lebesgue measurability or Baire property, which both follow from PD as well, it is natural to ask whether the Π^1_3-uniformization property bears large cardinal strength as well. We shall answer it negatively.

Theorem. *There is a generic extension of L in which the Π^1_3-uniformization property is true.*

The same uniformization construction also applies over canonical inner models with Woodin cardinals. This can be used to obtain better lower bounds in terms of consistency strength for the Π^1_n uniformization property for odd n. For even n we can produce for the first time models where the Π^1_n uniformization property holds true.

Theorem. *Let M_n be the canonical inner model with n Woodin cardinals. Then there is a generic extension of M_n in which the Π^1_{n+3} uniformization property holds true.*

Since the first version of this paper was written, the method developed here has led to several further constructions. A variant combines Π-uniformization with a projective well-order of the reals under CH [7]; another combines global Σ-uniformization with a large continuum and a projective well-order [8]; and the Σ-side of the local Delfino problem combines related coding methods with projective regularity and uniformization [9]. These later papers should be viewed as refinements and applications of the forcing machinery introduced here. The present paper isolates the original fixed-point and allowability mechanism needed to force projective Π-uniformization without assuming the corresponding determinacy strength.

Questions concerning the forceability of (local) consequences of PD do have a long tradition in set theory. There is a vast body of literature concerning the forcability of local levels of the projective hierarchy satisfying (Boolean combinations of) the Baire property, the perfect set property or Lebesgue measurability; see for instance [5]. There has been very little progress in the past, however, concerning similar questions for the separation,

the reduction and the uniformization property. Indeed, even the question of whether one can force the Σ^1_3-separation property, which is the weakest of said properties, remained an open problem for 50 years (see [15], Problem 3029).

This article continues this line of research and develops the forcing construction used in the subsequent work mentioned above. It is organized as follows: in the preliminaries section, we briefly introduce the forcings which we will use in the proof and produce a generic extension W of L which will be a well-suited ground model for our needs. We then start to prove the theorems from above. The main idea is to turn the problem of finding a partial order $\mathbb{P}$ which forces the Π^1_3-property into a fixed point problem. We shall define a derivation operator which acts on a specific set of forcing iterations of length ω_1. This operator will be applied transfinitely often, and will produce better and better approximations to the set of forcings we actually want to use in the end. The process is shown to converge in that eventually a fixed point $\mathcal{P}$, i.e. a suitable set of forcings is reached. Forcings which belong to this fixed point $\mathcal{P}$ allow for a certain, seemingly self-referential definition of an ω_1-length iteration which forces the Π^1_3-uniformization property over W. We then follow up, to alter the said process such that it becomes applicable to the canonical inner models M_n with n Woodin cardinals.

There are some similarities to [6], in particular the two proofs rely on a similar ground model W, which is a generic extension of L, and use a similar coding method which relies on a suitably chosen ω_1-sequence of ω_1-Suslin trees. However a more straightforward application of the ideas of [6] will fail badly to produce a model of the Π^1_3-uniformization property. As a consequence, a solution has to necessarily introduce several new ideas in order to succeed. The presentation of those is the goal of this paper.

2 Preliminaries

2.1 Notation

The notation we use will be mostly standard, we hope. We write $\mathbb{P} = (\mathbb{P}_\alpha : \alpha < \gamma)$ for a forcing iteration of length γ with initial segments $\mathbb{P}_\alpha$. The α-th factor of the iteration will be denoted with $\mathbb{P}(\alpha)$. Note here that we drop the dot on $\mathbb{P}(\alpha)$, even though $\mathbb{P}(\alpha)$ is in fact a $\mathbb{P}_\alpha$-name of a partial order.

2.2 The forcings which are used

The forcing notions used in the construction are standard. The preparatory extension is obtained solely by adding branches through a fixed sequence of Suslin trees, and the later coding uses the ground-model version of the ω_1-Cohen forcing together with Jensen–Solovay almost disjoint coding [11].

Recall that a Suslin tree is a normal tree of height ω_1 with countable levels and without an uncountable chain or antichain. Whenever S is a Suslin tree, we also write S for the forcing whose conditions are the nodes of S, ordered by end-extension. Thus forcing with S adds a cofinal branch through S.

Definition 2.1. *Let $\vec{S} = \langle S_\xi \mid \xi < \kappa \rangle$ be a sequence of Suslin trees. We call $\vec{S}$* independent *if, for every nonempty finite set $e \subseteq \kappa$, the product*

$$\prod_{\xi \in e} S_\xi$$

is again a Suslin tree.

For $A \subseteq \kappa$ let

$$\mathrm{Br}_A(\vec{S}) = \prod_{\xi \in A}^{\mathrm{fs}} S_\xi$$

denote the finite-support product of the corresponding branch forcings. When the sequence is fixed, we simply write Br_A.

Lemma 2.2. *Let $\vec{S} = \langle S_\xi \mid \xi < \kappa \rangle$ be an independent sequence of Suslin trees, where $\kappa \leq \omega_1$.*

1. *For every $A \subseteq \kappa$, the forcing Br_A is c.c.c.*
2. *If $A \subseteq B \subseteq \kappa$, then Br_A is a complete subforcing of Br_B, and*

$$\mathrm{Br}_B \cong \mathrm{Br}_A \times \mathrm{Br}_{B \setminus A}\,.$$

3. *If $\xi \notin A$, then*

$$\mathrm{Br}_A \Vdash \text{“}S_\xi \text{ is a Suslin tree.”}$$

Proof. For (1), let $\{p_\alpha \mid \alpha < \omega_1\}$ be a family of conditions in Br_A. After thinning it out, the supports form a Δ-system with finite root r, and the restrictions $p_\alpha \restriction r$ belong to the finite product $\prod_{\xi \in r} S_\xi$. By independence this finite product is Suslin, hence c.c.c., so two of the root restrictions are compatible. Since the remaining parts of their supports are disjoint, the corresponding two conditions in Br_A are compatible.

The factorization in (2) is the usual coordinatewise factorization of a finite-support product. For (3), observe that

$$\mathrm{Br}_A \times S_\xi \cong \mathrm{Br}_{A \cup \{\xi\}},$$

which is c.c.c. by (1). The standard product characterization of preservation of a Suslin tree now yields the assertion. □

We next recall the almost disjoint coding forcing due to Jensen and Solovay [11]. We identify subsets of ω with their characteristic functions. Let $D = \langle d_\alpha \mid \alpha < \omega_1 \rangle$ be an almost disjoint family of infinite subsets of ω, and let $X \subseteq \omega_1$.

Definition 2.3. *The almost disjoint coding forcing $\mathbb{A}_D(X)$ consists of pairs $(r, R) \in [\omega]^{<\omega} \times [D]^{<\omega}$. We put $(s, S) \leq (r, R)$ if and only if*

1. *$r \subseteq s$ and $R \subseteq S$;*
2. *whenever $\alpha \in X$ and $d_\alpha \in R$, then $s \cap d_\alpha = r \cap d_\alpha$.*

If $G \subseteq \mathbb{A}_D(X)$ is generic and $x_G = \bigcup\{r : (r, R) \in G\}$, then

$$\alpha \in X \quad \Longleftrightarrow \quad x_G \cap d_\alpha \text{ is finite.}$$

Fix a recursive injection $\# : 2^{<\omega} \to \omega$. If $\langle x_\xi : \xi < \omega_1^L \rangle$ enumerates the constructible reals in the canonical order, use $d_\xi = \{\#(x_\xi \restriction k) : k < \omega\}$ and $D = \langle d_\xi : \xi < \omega_1^L \rangle$.

Lemma 2.4. *For every $X \subseteq \omega_1$, the forcing $\mathbb{A}_D(X)$ is Knaster. In particular it is c.c.c. and preserves every ground-model Suslin tree.*

Proof. The Knaster property follows from the usual Δ-system argument applied to the finite side conditions. If T is a Suslin tree, then the product of a Knaster forcing with the c.c.c. forcing T is c.c.c. Hence $\mathbb{A}_D(X) \times T$ is c.c.c., which is equivalent to preservation of T as a Suslin tree. □

Finally, we write

$$\mathbb{C}_L(\omega_1) = ({}^{<\omega_1}2)^L$$

for the L-computed ω_1-Cohen forcing, ordered by end-extension. Thus conditions are countable binary sequences belonging to L. Whenever this forcing is used in an outer model of L, the notation refers to this fixed L-poset rather than to a recomputation in the ambient universe.

2.3 The ground model W of the iteration

We now define the ground model over which the main iteration will be carried out. We start with the canonical independent Suslin sequence already present in L and add a cofinal generic branch through every member of that sequence.

The following lemma gives the simultaneous form of Jensen's diamond construction which we use throughout the paper; see also [10]. We use the levelwise product of finitely many trees. This is a dense suborder of the corresponding finite forcing product.

Lemma 2.5. *There is a canonical L-definable sequence*

$$\vec{S} = \langle S_\xi \mid \xi < \omega_1 \rangle$$

of independent normal Suslin trees. The construction is given by a recursion which is uniform over L_{ω_1}.

Proof. Work in L. Fix the canonical L-diamond sequence $\langle a_\alpha \mid \alpha < \omega_1 \rangle$ and the canonical L-coding of finite sets of ordinals, countable trees, and subsets of their finite products. We recursively construct coherent countable approximations $\vec{S}^{\,\alpha} = \langle S^\alpha_\xi \mid \xi < \alpha \rangle$ such that each S^α_ξ is a normal splitting tree of height α and $S^\beta_\xi = S^\alpha_\xi \restriction \beta$ whenever $\xi < \beta < \alpha$. At limit stages we take unions.

It remains to describe the passage from α to $\alpha + 1$. For old coordinates $\xi < \alpha$, successor stages are handled by putting countably many immediate successors above each node on the top level. Suppose now that α is a nonzero limit. If a_α does not decode as a pair (e, A), where $e \in [\alpha]^{<\omega} \setminus \{\varnothing\}$ and A is a maximal antichain in the product

$$P^\alpha_e = \bigcup_{\delta<\alpha} \prod_{\xi \in e} (S^\alpha_\xi)_\delta,$$

then we use the $<_L$-least countable family of cofinal branches through each old tree which covers all old nodes, and put one new top node above each of these branches.

If a_α does decode such a pair (e, A), we choose the branch families for the coordinates in e simultaneously so as to seal A. Concretely, by a countable genericity argument in $\prod^{\mathrm{fs}}_{(\xi,n)\in e\times\omega} S^\alpha_\xi$, choose cofinal branches $b_{\xi,n}$, $\xi \in e$, $n < \omega$, such that they cover all old nodes and such that, for every $\nu : e \to \omega$, the tuple of new top nodes placed above $\langle b_{\xi,\nu(\xi)} \mid \xi \in e \rangle$ extends some member of A. The required dense sets are just the usual ones ensuring cofinality, coverage of old nodes, and the displayed sealing requirement; density of the last kind uses maximality of A in P^α_e. For $\xi \notin e$ we again use the default $<_L$-least covering family. Finally, the new coordinate α is initialized as the canonical finite-support binary tree of height $\alpha + 1$. This completes the recursion. Put $S_\xi = \bigcup_{\xi<\alpha<\omega_1} S^\alpha_\xi$. Then each S_ξ is a normal splitting tree of height ω_1 with countable levels.

We show that the sequence is independent. Fix a finite nonempty $e \subseteq \omega_1$ and let

$$P_e = \bigcup_{\alpha<\omega_1} \prod_{\xi \in e} (S_\xi)_\alpha$$

be the product. This is dense in the usual finite product of the corresponding tree forcings. Let $A \subseteq P_e$ be a maximal antichain in L. There is a club of $\alpha > \sup(e)$ such that $A \cap (P_e \restriction \alpha)$ is maximal in $P_e \restriction \alpha$, and, by coherence of the coding, a club on which the fixed code $x_{e,A}$ for (e, A) restricts to a code

for $(e, A \cap (P_e \restriction \alpha))$. By $\diamondsuit$, choose a limit α in the intersection of these clubs with $a_\alpha = x_{e,A} \cap \alpha$. At stage α the construction seals $A \cap (P_e \restriction \alpha)$. Hence every node on level α, and therefore every later node, extends a member of this initial antichain. Since A itself is an antichain, $A = A \cap (P_e \restriction \alpha)$, so A is countable. Thus P_e has no uncountable antichain. A cofinal branch through P_e would, by normal splitting, yield an uncountable antichain by choosing at successor levels nodes incompatible with the branch. Hence P_e is Suslin. Since e was arbitrary, $\vec{S}$ is independent. The construction is canonical from $<_L$, the diamond sequence, and the fixed coding, and is therefore uniform over L_{ω_1}. □

Lemma 2.6. *Let M be a transitive uncountable model of*

$$\mathsf{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$$

such that

$$\omega_1^M = \omega_1^L = \omega_1.$$

There is a Σ_1-formula

$$\vartheta_{\mathrm{Sus}}(t, \gamma, \eta, \omega_1),$$

independent of M, which defines over M the relation

$$\{(t, \gamma, \eta) : \gamma, \eta < \omega_1 \text{ and } t \in (S_\eta)_\gamma\}.$$

Equivalently, there is a formula $\mathrm{CanSuslin}_L(X, \omega_1)$, which may be taken to be $\Sigma_1(\omega_1)$ over $H(\omega_2)$, such that in every ω_1-preserving outer model U of L,

$$U \models \mathrm{CanSuslin}_L(X, \omega_1) \quad \Longleftrightarrow \quad X = \vec{S}.$$

In particular, the canonical sequence $\vec{S}$ is computed in the same way in all the outer models considered below.

Proof. By Lemma 2.5, the sequence $\vec{S}$ is obtained by a canonical recursion over L_{ω_1}, and the corresponding level relation is $\Sigma_1(\omega_1)$-definable over L_{ω_1}. If M is transitive, satisfies ZFC^- + “$\aleph_1$ exists”, and has $\omega_1^M = \omega_1^L$, then M computes L_{ω_1} correctly. Hence M computes the same canonical diamond sequence, the same coding apparatus, and therefore the same recursive construction of $\vec{S}$. The same Σ_1-formula $\vartheta_{\mathrm{Sus}}(t, \gamma, \eta, \omega_1)$ consequently defines over M the relation $t \in (S_\eta)_\gamma$.

The formulation by $\mathrm{CanSuslin}_L(X, \omega_1)$ is obtained by saying that X is the union of the unique system produced by this canonical recursion. Since every ω_1-preserving outer model of L has the same L_{ω_1}, it satisfies this formula exactly for the true sequence $\vec{S}$. □

Let

$$\mathrm{Br} = \prod_{\xi < \omega_1}^{\mathrm{fs}} S_\xi$$

be the finite-support branch product, computed in L, and let $H \subseteq \mathrm{Br}$ be L-generic. For every $\xi < \omega_1$ set

$$b_\xi = \bigcup \{p(\xi) : p \in H \text{ and } \xi \in \mathrm{dom}(p)\}.$$

Then b_ξ is a cofinal L-generic branch through S_ξ. In this paper, to *destroy* a Suslin tree always means to add such a cofinal generic branch through it. We define

$$W = L[H].$$

Theorem 2.7. *The model W has the following properties.*

1. Br *is a c.c.c. forcing of size* $\aleph_1$*. Hence W preserves all cardinals and cofinalities of L, and $W \models \mathsf{CH}$.*
2. *Every S_ξ is destroyed in W, witnessed by the branch b_ξ.*
3. *For $A \subseteq \omega_1$ in L, let $H_A = H \cap \mathrm{Br}_A$. Then H_A is Br_A-generic over L, and*
$$L[H_A] \models \text{“}S_\xi \text{ has a cofinal branch”} \quad \iff \quad \xi \in A.$$
4. *Every real in W belongs to $L[H_A]$ for some countable $A \subseteq \omega_1$ belonging to L.*

Proof. The first assertion follows from Lemma 2.2. Since $L \models \mathsf{CH}$ and Br has size $\aleph_1$, there are only $\aleph_1$ many nice Br-names for reals, and therefore W still satisfies CH. The definition of H and the dense sets requiring a condition to reach a prescribed level of a prescribed coordinate give (2).

For (3), use the factorization

$$\mathrm{Br} \cong \mathrm{Br}_A \times \mathrm{Br}_{\omega_1 \setminus A}.$$

If $\xi \in A$, then $b_\xi \in L[H_A]$. If $\xi \notin A$, then Lemma 2.2 shows that Br_A preserves S_ξ as a Suslin tree, so no cofinal branch through S_ξ belongs to $L[H_A]$.

Finally, let $\dot{x}$ be a nice Br-name for a real. The antichains occurring in $\dot{x}$ are countable, and the union of the finite supports of all conditions appearing in them is therefore a countable set $A \subseteq \omega_1$ in L. Thus $\dot{x}$ is already a Br_A-name, proving (4). □

The relevant inner models of W used below are obtained by retaining selected branch coordinates from H.

Coding forcing. The extensions under consideration preserve ω_1^L, so $\omega_1 = \omega_1^L$. Let $u \in 2^\omega$ and let g be the generic subset of ω_1 at a fresh $\mathbb{C}_L(\omega_1)$ coordinate. Let $\varrho_L : ({}^{<\omega_1}2)^L \to \omega_1$ be the canonical bijection. Put

$$h(g) = \{\varrho_L(g \restriction \gamma) : \gamma < \omega_1\},$$
$$I(u, g) = \{\omega\gamma + 2k + u(k) : \gamma \in h(g),\ k < \omega\}.$$

Let $X \subseteq \omega_1$ canonically code $(u, h(g), \langle b_\xi : \xi \in I(u, g)\rangle)$. We describe the localization $Y = Y(X)$ used by $\mathrm{Code}(u, \eta)$, where η specifies the coordinate supplying g.

Use a recursive theory T consisting of ZFC^- with Collection and Choice, existence of ω_1, the relativizations of these axioms to $L[a]$ for each real a, and $\mathrm{HC}^L = L_{\omega_1^L}$. For a transitive $N \models T$ with $u \in N$ and $\omega_1^N = (\omega_1^L)^N = \nu$, put

$$\mathrm{Pat}(N, u, \nu) \quad \iff \quad N \models \forall \beta < \nu\ \exists \gamma \in [\beta, \nu)\ \ \mathrm{Pattern}(\gamma, u).$$

Here $\mathrm{Pattern}(\gamma, u)$ asserts that for every $k < \omega$ there is a cofinal branch through $S_{\omega\gamma+2k+u(k)}$. The predicate includes the displayed requirements on N, u, ν. The trees are defined by Lemma 2.6.

In $L[X]$, collapse an elementary submodel of a sufficiently large H_χ of size $\aleph_1$ containing X, u, ω_1 and all ordinals below ω_1. Its collapse N satisfies $\mathrm{Pat}(N, u, \omega_1)$. Choose a bijection $e : \omega_1 \to N$ and code membership and satisfaction via e by $Z \subseteq \omega_1$, marking the indices of u and ω_1. Use finite tuple codes ordered by maximum. In $L[Z]$, take the club of closure points ν at which the code restricts correctly and $e[\nu] \prec N$, with $u, \omega_1 \in e[\nu]$ and $e[\nu] \cap \omega_1 = \nu$. Such a club is obtained by closing under the Skolem maps, e and its inverse. At these points the collapse N_ν satisfies $\mathrm{Pat}(N_\nu, u, \nu)$. Let E consist of the nonzero limit points of this club and 0. If $c \in E$, write $d_c = \min(E \setminus (c + 1))$ and let w_c be the $<_L$-least real coding a well-order of type d_c. Set

$$Y(X) = \{2 \cdot \xi + 1 : \xi \in Z\} \cup \{2 \cdot (c + k) : c \in E,\ k < \omega,\ w_c(k) = 1\}.$$

The intervals $[c, c + \omega)$ are disjoint. Make the choices in $L[X]$ and $L[Z]$ by their canonical well-orders, so the same X gives the same Y in models with the correct ω_1. This is the localization argument of David [3]. The second factor of $\mathrm{Code}(u, \eta)$ is $\mathbb{A}_D(Y(X))$.

Let $\Psi(r, u)$ assert that for every countable transitive $M \models T$ with $r, u \in M$ and $\omega_1^M = (\omega_1^L)^M$, one has

$$M \models \exists N \in L[r]\ \ \mathrm{Pat}(N, u, \omega_1^M).$$

A real coding M can also code its satisfaction relation. The well-foundedness requirement is Π^1_1 and the remaining tests on the code are arithmetic. Hence Ψ is Π^1_2, and $\Phi(u) \iff \exists r\, \Psi(r, u)$ is Σ^1_3.

Lemma 2.8. *The generic real r of the almost disjoint factor of* $\mathrm{Code}(u,\eta)$ *satisfies* $\Psi(r,u)$*. Conversely, in an extension preserving* ω_1*, if* $\Psi(r,u)$ *holds, then for unboundedly many* $\gamma<\omega_1$ *and every* $k<\omega$*,* $L[r]$ *contains a cofinal branch through* $S_{\omega\gamma+2k+u(k)}$*.*

Proof. Given a countable transitive M as in the definition of Ψ, put $\nu=\omega_1^M$. The definition of D and $\mathrm{HC}^L=L_{\omega_1^L}$ give the correct $D\restriction\nu$ in M. The almost disjoint real decodes $Y\cap\nu$ in $L[r]^M$. If $\nu\notin\mathrm{Lim}(E)$, let $c=\max(E\cap\nu)$. Then $c+\omega<\nu\le d_c$. The even part gives $w_c\in M$, contrary to $\nu=\omega_1^M$. At a limit point of E the odd part yields the code of N_ν, so the collapse lies in $L[r]^M$ and is as required.

For the converse, use $F=H_\chi^{L[r,u]}$ for sufficiently large regular χ. This model satisfies T and has $\omega_1^F=(\omega_1^L)^F=\omega_1$. If the conclusion in the definition of Ψ failed at F, a countable elementary submodel of F containing r,u,ω_1 would collapse to a countable transitive model witnessing failure. Thus F sees such an $N\in L[r]$. The trees at ω_1 are computed correctly, and the branches in N belong to $L[r]$. Finally, once r witnesses $\Phi(u)$ it continues to witness it by Shoenfield absoluteness. □

2.4 Coding iterations and their initial segments

For $E\in L$ of size at most $\aleph_1$, put

$$C_E=\mathrm{Fn}_{<\omega_1}(E\times\omega_1,2)^L.$$

Use the Cohen index sets $E_i=[\omega_1\cdot i,\omega_1\cdot(i+1))$ and put $E_{<\lambda}=\bigcup_{i<\lambda}E_i$. At stage i we first force with C_{E_i} and then with an almost disjoint factor $\dot A_i$, which may be trivial. Thus P_i is the initial segment through the first i stages, and $P_i*C_{E_i}$ is the forcing just before $\dot A_i$.

Whether to code at stage i, the real u_i to be coded, the index $\eta_i\in E_i$, and the isomorphism $\varphi_i:C_{E_i}\cong C_{E_i}\restriction c_i$ are given by P_i-names. Each possible value of φ_i is in L and is induced by bijections on the individual Cohen coordinates, as in the proof below, with at most countably many nonidentity coordinates. If K_i is C_{E_i}-generic, let g_i be the η_i-th Cohen subset of ω_1 obtained from the filter generated by $\varphi_i[K_i]$. At a coding stage put

$$I_i=I(u_i,g_i),\qquad X_i=\mathrm{code}(u_i,h(g_i),\langle b_\xi:\xi\in I_i\rangle)$$

and use $\mathbb{A}_D(Y(X_i))$, possibly restricted below a condition a_i. The name for a_i is a $P_i*C_{E_i}$-name. At other stages we still use C_{E_i}, but $\dot A_i$ is trivial and $I_i=\varnothing$.

For $\lambda\le\omega_1$, write the forcing as

$$P_\lambda=C_{E_{<\lambda}}*\dot R_\lambda,\qquad R_\lambda=\mathop{{\ast}}_{i<\lambda}^{\mathrm{fs}}\dot A_i. \tag{1}$$

Here the names for $\dot{A}_i$ are translated from $P_i * C_{E_i}$. The later Cohen forcings are ground-model partial orders and the names for the earlier factors use only the earlier Cohen coordinates. Moreover the order on $\mathbb{A}_D(Y) \restriction a$ is determined by Y and a. Thus the later Cohen forcings can be moved before the earlier almost disjoint factors. At a limit, (1) uses the countable-support Cohen product computed in L and the finite-support almost disjoint iteration. For $\lambda = \omega_1$, the Cohen product is isomorphic to the full reservoir.

Use the dense presentation of R_λ in which a condition is a finite function f assigning pairs in $[\omega]^{<\omega} \times [\omega_1]^{<\omega}$ to stages in its support. An omitted stage denotes the greatest condition of its factor. At a successor one strengthens the preceding condition to decide the name for the next pair. At a limit a finite support is bounded. This proves density by induction. The same induction shows that the presentations agree in two nested models with the same ordinals if the factors have the same conditions and order in common generic extensions by the preceding factors. At a successor, opposite forcing decisions about membership or order of a pair would contradict agreement in a generic extension of the larger model. At a limit each such decision is made in a bounded initial segment.

We retain the complete embeddings of initial segments and the corresponding translations of names as part of the presentation. A tail in this presentation has a fixed countable length and uses Cohen coordinates disjoint from those of the preceding iteration. The bijections of Cohen index sets used to present products and quotients belong to L.

Lemma 2.9. *The form* (1) *is preserved by the following operations.*

1. *For $\beta \leq \lambda$ and $G_\beta \subseteq P_\beta$ generic,*

$$P_\beta \lessdot P_\lambda, \qquad P_\lambda / G_\beta \simeq C_{E_{<\lambda} \setminus E_{<\beta}} * \dot{R}_{[\beta,\lambda)}.$$

   On the dense presentation the projection is

$$\pi_\beta(c, f) = (c \restriction (E_{<\beta} \times \omega_1), f \restriction \beta).$$

2. *A tail defined below $0 < d \in \mathrm{RO}(P)$ lifts to a P-name by retaining its Cohen factors and taking*

$$\dot{A}_i^{\uparrow} = \begin{cases} \dot{A}_i, & \text{on } d, \\ \mathbf{1}, & \text{on } \neg d. \end{cases}$$

   For p in the dense presentation, $P_\lambda \restriction p$ again has form (1)*, with initial segments $P_\beta \restriction \pi_\beta(p)$, $\beta \leq \lambda$.*

3. *Tails Q, R over the same preceding forcing admit product presentations in either order, on disjoint Cohen index sets. With R first, $R \times Q_i$ is an initial segment, and*

$$R \times Q_i \lessdot R \times Q.$$

If $\dot{T}$ is a Q_i-name for a continuation of fixed countable length, lifted by (2) if necessary, its translation $\dot{T}_R$ satisfies

$$(R \times Q_i) * \dot{T}_R \simeq R \times (Q_i * \dot{T})$$

over $R \times Q_i$, preserving all previously specified embeddings and translations of names.

Proof. (1) Use

$$C_{E_{<\lambda}} \cong C_{E_{<\beta}} \times C_{E_{<\lambda} \setminus E_{<\beta}}.$$

If $p = (c, f)$ and $q = (c', f') \leq \pi_\beta(p)$, then

$$\big(c' \cup c \restriction ((E_{<\lambda} \setminus E_{<\beta}) \times \omega_1),\ f' \cup f \restriction [\beta, \lambda)\big) \leq p, q.$$

The later values of f remain legal after strengthening the earlier conditions. This proves completeness and the quotient formula, also at limits.

(2) The lift is Boolean mixing. For the restriction below (c, f), choose in L, for each $\eta \in \operatorname{supp}(c)$,

$$\sup(\operatorname{dom} c_\eta) + \omega < \nu_\eta < \omega_1,$$
$$\tau_\eta : \omega_1 \longrightarrow \omega_1 \setminus \operatorname{dom} c_\eta \quad \text{a bijection,} \qquad \tau_\eta \restriction [\nu_\eta, \omega_1) = \mathrm{id}.$$

Set $\tau_\eta = \mathrm{id}$ elsewhere. Then

$$\psi_c : C_E \cong C_E \restriction c, \qquad \psi_c(q)_\eta = c_\eta \cup (q_\eta \circ \tau_\eta^{-1}).$$

Compose the block maps with φ_i and replace A_i by $A_i \restriction f(i)$ for $i \in \operatorname{dom}(f)$. Below (c, f) this preserves $g_i, X_i, Y(X_i)$. By (1) the maps commute with all initial-segment projections; their bounded coordinate changes are closed under composition.

(3) Reindex the Cohen coordinates disjointly in L and translate each tail's names from its own initial segments. In either order the almost disjoint factors have identical conditions and orders, so induction on the dense presentations gives the product isomorphisms. For a continuation of $R \times Q_i$, retain both index maps, extend the Q_i-map, and translate the Boolean value in (2) by the fixed embedding of Q_i. This gives the displayed isomorphism over $R \times Q_i$. The same induction applies to translations along an isomorphism and its inverse. □

Lemma 2.10. *For $\lambda \leq \omega_1$,*

$$\operatorname{dens}(P_\lambda) \leq \aleph_1, \qquad P_\lambda \text{ preserves cardinals and } \mathsf{CH}.$$

If $G_{\omega_1} \subseteq P_{\omega_1}$ is generic and $G_\beta = G_{\omega_1} \cap P_\beta$, then

$$(2^\omega)^{W[G_{\omega_1}]} = \bigcup_{\beta < \omega_1} (2^\omega)^{W[G_\beta]}.$$

Proof. Put $C = C_{E_{<\lambda}}$ and let $K \subseteq C$ be generic over W. Since $L \models |C| \leq \aleph_1$ and C is countably closed in L, Easton's lemma applied to Br gives

$$(\mathrm{Ord}^\omega)^{W[K]} = (\mathrm{Ord}^\omega)^W.$$

Thus C preserves cardinals and CH over W. In $W[K]$, R_λ is c.c.c. with a dense set of finite functions of size $\leq \aleph_1$. Hence $C * \dot{R}_\lambda$ has the asserted properties.

For $z \in (2^\omega)^{W[G_{\omega_1}]}$, take a nice R_{ω_1}-name $\dot{z}$ in the dense presentation. Its ordinal code belongs to W, and

$$\mathrm{supp}(\dot{z}) = \bigcup\{\mathrm{dom}(p) : (\check{n}, p) \in \dot{z}\} \subseteq \beta \quad \text{for some } \beta < \omega_1.$$

The dense presentations of R_β in $W[K \restriction E_{<\beta}]$ and $W[K]$ agree. Thus $\dot{z}$ is already a name over P_β, and $z \in W[G_\beta]$. □

Lemma 2.11. *In every extension by an iteration of the form* (1)*,*

$$\{u \in 2^\omega : \Phi(u)\} = \{u_i : i < \lambda,\ i \text{ is a coding stage}\}.$$

Proof. The inclusion from right to left follows from Lemma 2.8, also for factors restricted below conditions. Conversely, let r witness $\Phi(u)$. Work first in $W[K]$, with $K \subseteq C_{E_{<\lambda}}$ generic, and let $G \subseteq R_\lambda$ be the remaining generic.

Choose names for r, u and the stage parameters using countable antichains in the dense presentation. Let $D_i \subseteq i$ collect the supports of the parameter names at stage i, including a_i. Put

$$J_0 = \mathrm{supp}(\dot{r}) \cup \mathrm{supp}(\dot{u}),$$
$$J_{n+1} = J_n \cup \bigcup_{i \in J_n} D_i, \qquad J = \bigcup_{n<\omega} J_n \in [\lambda]^{\leq\omega}.$$

The retained names have a common countable ordinal code z. By Easton's lemma and the c.c.c. of Br,

$$z \in W, \qquad z \in L[H_B] \quad \text{for some } B \in [\omega_1]^{\leq\omega} \cap L.$$

Since $D_i \subseteq J \cap i$ for $i \in J$, induction gives

$$R_{J\cap i} \lessdot R_i \quad (i \leq \lambda), \qquad \dot{r}^{G_J} = r, \quad \dot{u}^{G_J} = u, \quad G_J = G \cap R_J.$$

At successors the retained factor is independent of the quotient; at limits use finite supports and the coherent projections.

Fix $\xi \notin B \cup \bigcup_{i\in J} I_i$ and put

$$M_\xi = L[K][H \restriction (\omega_1 \setminus \{\xi\})].$$

Then $z \in M_\xi$. In M_ξ form $\widetilde{R}^\xi_J$ by following R_J until the first stage whose names fail to determine unique data or whose I_i contains ξ, and taking trivial factors from that stage onward. The test $\xi \in I_i$ precedes the formation of X_i, so the retained factors have identical dense presentations over M_ξ and $W[K]$. In $W[K]$, induction gives

$$d_\xi = \bigwedge_{i \in J} \|\xi \notin \dot{I}_i\|_{\mathrm{RO}(R_J)},$$

$$\widetilde{R}^\xi_J \lessdot R_J, \qquad \widetilde{R}^\xi_J \restriction d_\xi \simeq R_J \restriction d_\xi.$$

At limits take meets. Our choice of ξ makes G_J meet $R_J \restriction d_\xi$, inducing an M_ξ-generic $\widetilde{G} \subseteq \widetilde{R}^\xi_J$ with

$$r, u \in M_\xi[\widetilde{G}].$$

The Cohen product preserves the c.c.c. of ground-model branch products, whence

$$L[K] \models \mathrm{Br}_{\omega_1 \setminus \{\xi\}} \times S_\xi \text{ is c.c.c.}$$

All nontrivial factors of $\widetilde{R}^\xi_J$ are σ-centered. Hence

$$M_\xi[\widetilde{G}] \models \text{“}S_\xi \text{ is Suslin.”}$$

Let $\mathcal{C} = \{i \in J : i \text{ is a coding stage}\}$. Distinct Cohen coordinates disagree unboundedly, and the coordinate isomorphisms change only bounded initial segments. Thus their common initial segments have bounded length, and

$$\sup\big(h(g_i) \cap h(g_j)\big) < \omega_1 \qquad (i \neq j \in \mathcal{C}).$$

Suppose no stage codes u. Choose $\delta < \omega_1$ with

$$B \subseteq \delta, \qquad h(g_i) \cap h(g_j) \subseteq \delta \quad (i \neq j \in \mathcal{C}).$$

By Lemma 2.8, choose $\gamma > \delta$ such that

$$\forall k < \omega \quad L[r] \models \text{“}S_{\omega\gamma+2k+u(k)} \text{ has a cofinal branch.”}$$

There is at most one $i \in \mathcal{C}$ with $\gamma \in h(g_i)$. If such i exists, choose k with $u_i(k) \neq u(k)$; otherwise put $k = 0$. Then

$$\xi = \omega\gamma + 2k + u(k) \notin B \cup \bigcup_{i \in J} I_i.$$

But $L[r] \subseteq M_\xi[\widetilde{G}]$, contradicting the Suslinity of S_ξ in that model. □

Up to the specified presentations, the bounded iterations, their restrictions below conditions, and the names for their tails form a set. Each dense set can be indexed by a subset of ω_1, each Boolean value is represented by a subset of that dense set, and the names for the parameters at the stages and for the complete embeddings can be given by Boolean-valued functions. For a tail whose countable length is given by a name, first work below a condition deciding the length and apply Lemma 2.9(2). Thus the Boolean values used below may be indexed by tail names of fixed countable length.

2.5 Motivation for the allowability hierarchy

Given the notion of allowable forcing, we now form successive approximations to the class of forcings which we eventually want to use. We first describe the idea leading to these approximations.

Fix a recursive enumeration

$$\langle \varphi_m(v_0, v_1) \mid m \in \omega \rangle$$

of the Π^1_3-formulas with two free real variables, and put

$$A_m := \{(x, y) \in (2^\omega)^2 : \varphi_m(x, y)\}.$$

We write f_m for the uniformizing function which we intend to define for A_m, and write $f(m, x)$ for $f_m(x)$. The goal is to choose, for every $m \in \omega$ and every real x for which

$$A_{m,x} := \{y \in 2^\omega : (x, y) \in A_m\}$$

is nonempty, a real $f(m, x)$ such that

$$(x, f(m, x)) \in A_m,$$

and to code every triple (x, y, m) such that

$$y \neq f(m, x) \quad \text{and} \quad (x, y) \in A_m$$

into $\vec{S}$. The final forcing will have length ω_1, and each of its countable initial segments will be allowable. Since the statement

$$\text{“}(x, y, m) \text{ is coded into } \vec{S}\text{”}$$

is Σ^1_3, the requirement

$$(x, y) \in A_m \quad \text{and} \quad (x, y, m) \text{ is not coded into } \vec{S}$$

is Π^1_3. The bookkeeping of the final iteration will ensure that, whenever $A_{m,x} \neq \varnothing$, exactly one such y remains.

The idea behind the definition can be seen in the following toy example. Work in W, let $x \in W$, and suppose that

$$A_{m,x} = \{y_0, y_1\}.$$

Assume in addition that no allowable forcing over W adds a new member to $A_{m,x}$. We consider only the problem of defining $f(m, x)$ and omit, for the moment, the interaction with the other sections.

Call $y \in A_{m,x}$ *eliminable* if there is an allowable forcing $\mathbb{P}$ such that

$$\mathbb{P} \Vdash (x, y) \notin A_m.$$

Call y *persistent* if it is not eliminable. Thus, if y_0 is persistent, we may reserve y_0 as the value of $f(m,x)$ and code (x,y_1,m) into $\vec{S}$. If y_0 is eliminable, witnessed by an allowable forcing $\mathbb{P}$, we consider y_1. If y_1 is persistent, we reserve y_1 and code (x,y_0,m).

It remains to consider the case in which both values are eliminable. Let $\mathbb{P}$ and $\mathbb{Q}$ witness, respectively, that

$$\mathbb{P} \Vdash (x,y_0) \notin A_m \qquad \text{and} \qquad \mathbb{Q} \Vdash (x,y_1) \notin A_m.$$

By Lemma 2.9, the product

$$\mathbb{P} \times \mathbb{Q}$$

is isomorphic to an allowable forcing. Using this presentation of the product, upward absoluteness gives

$$\mathbb{P} \times \mathbb{Q} \Vdash (x,y_0) \notin A_m \wedge (x,y_1) \notin A_m.$$

By the assumption that no new member is added to the section, the product forces

$$A_{m,x} = \varnothing.$$

In this case there is no value to uniformize at x.

This argument isolates the two operations which are needed later:

reserve a value which is persistent under the forcings under consideration;
combine eliminating forcings by taking products. (2)

The difficulty is that an eliminating forcing may add codes which interfere with other sections. Thus it is not enough to use arbitrary allowable forcings as witnesses. We need a class $\mathcal{P}$ of allowable forcings such that the following rule produces forcings in $\mathcal{P}$ again. For every task (x,m) considered by the bookkeeping, either

$$\exists y \in A_{m,x}\ \forall \mathbb{Q} \in \mathcal{P}\big(\mathbb{Q} \text{ extending the current forcing} \to \mathbb{Q} \Vdash (x,y) \in A_m\big),$$

in which case one such y is reserved and the other values are coded, or every value under consideration can be eliminated by a forcing in $\mathcal{P}$.

This turns the construction into a fixed point problem. Let

$$\mathcal{A}_0^W$$

be the class of allowable iterations. The first refinement searches for values which are persistent under $\mathcal{A}_0^W$-extensions; the resulting iterations form $\mathcal{A}_1^W$. The next refinement searches also for values which are persistent under $\mathcal{A}_1^W$-extensions, and so on. We obtain a decreasing hierarchy

$$\mathcal{A}_0^W \supseteq \mathcal{A}_1^W \supseteq \cdots \supseteq \mathcal{A}_\alpha^W \supseteq \cdots .$$

In the first refinement we keep a minimizer uncoded whenever all values that can attain the least rank are already present.

The reason for continuing the construction is the usual moving-target problem. In the definition of 1-allowability we test persistence under 0-allowable extensions, but the iteration which we produce is itself 1-allowable. We should therefore also test persistence under 1-allowable extensions. Repeating this argument leads to the transfinite hierarchy defined in the next subsection.

2.6 Persistence and the choice of a uniformizing value

Use the coding iterations of Subsection 2.4, with the specified complete embeddings of their initial segments. We work in extensions M by such an initial segment, sometimes restricted below a condition. An iteration over M is a quotient with the presentation given in that subsection. The phrase *code* (m, x, y) means to code the real $\langle x, y, m\rangle$. Put

$$A(m, x, y) \quad \Longleftrightarrow \quad (x, y) \in A_m.$$

We use two absoluteness facts for A. A witness to $\neg A(b)$ in the extension by one factor of a product remains a witness in the product extension. If $\neg A(b)$ holds in the final extension, its witness gives $\neg A(b)$ in each sufficiently late bounded initial segment of the iteration. Here $b = (m, x, y)$ belongs to the model in which membership is tested. In the construction over L, write $\neg A(b)$ as $\exists z\, \psi(b, z)$ with $\psi \in \Pi^1_2$ and apply Shoenfield absoluteness to ψ. The second assertion also uses the fact that every real belongs to an extension by a bounded initial segment.

Choice functions. For each extension M under consideration we include a name for a choice function J_M on its nonempty sets of reals, so $J_M(B) \in B$. Along each specified inclusion $M \subseteq N$, require $J_N(B) = J_M(B)$ for $B \in \mathrm{dom}(J_M)$. These are names over the given forcings.

Lemma 2.12 (Choice functions in product extensions)**.** *Let Q, R be countable iterations over M. For their product extensions put*

$$M_{ij} = M[G_{Q_i}, G_{R_j}], \qquad \mathcal{D}_{ij} = \big(\mathcal{P}(\mathbb{R}) \setminus \{\varnothing\}\big)^{M_{ij}}.$$

Coherent choice functions on the two axes, agreeing at $M = M_{00}$, extend by names to functions

$$J_{ij} : \mathcal{D}_{ij} \longrightarrow \mathbb{R}^{M_{ij}}, \qquad J_{ij}(B) \in B,$$
$$i \le i',\ j \le j' \quad \Longrightarrow \quad J_{i'j'} \restriction \mathcal{D}_{ij} = J_{ij}.$$

Any prescribed coherent extension on $i \le i_0,\ j \le j_0$ may be retained. The assertions remain valid under restrictions below conditions, Boolean mixing, quotients by $Q_{i_0} \times R_{j_0}$, and appending an iteration to either factor of a specified product.

Proof. Order pairs coordinatewise, with meet $a \wedge b$ and join $a \vee b$. We first show

$$\mathcal{D}_a \cap \mathcal{D}_b = \mathcal{D}_{a \wedge b}.$$

For incomparable a, b, factor over $N = M_{a \wedge b}$. Fix a bijection $e : \theta \to \mathbb{R}^N$ in N. Generic intersection for the independent quotients, applied first to each element of B and then to $e^{-1}[B]$, gives

$$B \in \mathcal{D}_a \cap \mathcal{D}_b \implies B \subseteq \mathbb{R}^N \implies e^{-1}[B] \in N \implies B \in \mathcal{D}_{a \wedge b}.$$

The comparable case is immediate. Consequently

$$\ell(B) = \big(\min\{i : \exists j \ B \in \mathcal{D}_{ij}\}, \min\{j : \exists i \ B \in \mathcal{D}_{ij}\}\big)$$

satisfies

$$B \in \mathcal{D}_{\ell(B)}, \qquad B \in \mathcal{D}_a \iff \ell(B) \leq a.$$

Indeed, take the meet of pairs witnessing the two coordinate minima.

Let J^0_a denote the prescribed functions and let $\lhd_a$ be a named well-order of $\mathbb{R}^{M_a}$. For $B \in \mathcal{D}_a$, set

$$J_a(B) = \begin{cases} J^0_{\ell(B)}(B), & \text{if } \ell(B) \text{ is prescribed,} \\ \min_{\lhd_{\ell(B)}} B, & \text{otherwise.} \end{cases}$$

The prescribed region is downward closed, so all prescribed values are retained and $a \leq b \Rightarrow J_b \restriction \mathcal{D}_a = J_a$. These are names: for $a \leq b$, membership in the smaller extension is expressed by

$$\|\dot{B} \in M_a\| = \bigvee_{\dot{C}} \|\dot{B} = \iota_{ab}(\dot{C})\|,$$

where ι_{ab} is the specified translation and $\dot{C}$ ranges over a fixed set of names for sets of reals over $Q_{a_0} \times R_{a_1}$, where $a = (a_0, a_1)$. For $B \in \mathcal{D}_b$, the minima defining $\ell(B)$ can be computed using only pairs $a \leq b$.

If $B \in \mathcal{D}_c$ is old and occurs at a new pair a, then $B \in \mathcal{D}_{a \wedge c}$, with $a \wedge c$ old. Hence $\ell_{\text{new}}(B) = \ell_{\text{old}}(B)$. For $s = (i_0, j_0)$, restricting to pairs $a \geq s$ instead gives

$$\ell_{\geq s}(B) = \ell(B) \vee s.$$

On a boundary axis the choice is inherited; otherwise $\ell(B) \vee s = \ell(B)$, so the original choice applies. This gives compatibility with quotients by $Q_{i_0} \times R_{j_0}$ and with both product orders.

For restrictions use the specified embeddings and

$$(Q \times R) \restriction (q, r) = (Q \restriction q) \times (R \restriction r).$$

Below d retain the prescribed names; on $\neg d$ use a coherent extension and mix the names. Finally, for a continuation $\dot{T}$ over Q_{i_0} given below d, Lemma 2.9(2) supplies $\dot{T}^{\uparrow}$ with trivial almost disjoint factors off d. Apply the construction to $Q_{i_0} * \dot{T}^{\uparrow}$ and the other factor. The invariance of $\ell(B)$ retains the old choices, and Lemma 2.9(3) retains the specified embeddings. □

We work with the set of all countable iterations under consideration, together with their specified embeddings and the names for their choice functions. The same forcing may occur with different data. For a presentation of density at most $\aleph_1$, the regular open algebra has size at most $2^{\aleph_1}$, and names for reals are coded by countable sequences of its elements. A name for a set of reals gives the Boolean value of membership for each such real name. The same description, applied to the graph, gives names for the choice functions. Thus the presentations together with the names and embeddings still form a set, possibly of size greater than $2^{\aleph_1}$. This set includes the restrictions and products in Lemma 2.12. A name for a quotient iteration defined below a condition can be used there or extended by trivial almost disjoint factors on the complement, with the same countable length.

The allowability hierarchy. We define successive hierarchies of allowable iterations. Each hierarchy begins at a specified initial segment of the final iteration. The ξ-th hierarchy retains the restrictions on coding imposed so far and the fixed classes from hierarchies $\eta < \xi$. A restriction forbids coding a specified triple (m, x, y). At a later extension by a bounded initial segment, read the earlier definitions using the complete embeddings from the initial segments at which those definitions began. In a product with a further iteration these embeddings are obtained through the factor belonging to the final iteration. Only the construction of the final iteration begins a new hierarchy.

For an extension M considered in the current hierarchy, let $\mathcal{K}_0^\xi(M)$ consist of the quotient iterations over M that obey all the restrictions on coding and belong to every earlier fixed class, evaluated over M. Trivial almost disjoint factors are permitted. Given $\mathcal{K}_\beta^\xi$ in all these extensions, put

$$P_\beta^\xi(M) = \{b \in (\omega \times \mathbb{R}^2)^M : \forall Q \in \mathcal{K}_\beta^\xi(M) \quad Q \Vdash_M A(b)\}.$$

The rank through (ξ, β) of b is the first pair, in lexicographic order, at which b belongs to a persistence set. Use all levels of the earlier hierarchies up to their fixed levels and the levels at most β of the current hierarchy. Write $\rho_{\leq(\xi,\beta)}(b)$ for this partial rank. It is undefined precisely when b has no rank at any of these levels, including those in the earlier hierarchies.

For $m, x \in M$, let $\Lambda_\beta^\xi(M, m, x)$ be the set of pairs λ for which some $Q \in \mathcal{K}_\beta^\xi(M)$, some $q \in Q$, and some real name $\dot{y}$ satisfy

$$q \Vdash_Q^M \rho_{\leq(\xi,\beta)}(m, x, \dot{y}) = \lambda.$$

When this set is nonempty, put $\lambda = \min \Lambda_\beta^\xi(M, m, x)$. Call this minimum *stable* if no condition witnessing it forces $\dot{y} \notin \mathbb{R}^M$. In this case let B be the nonempty set of $y \in \mathbb{R}^M$ for which some such condition forces

$\rho_{\leq(\xi,\beta)}(m,x,\check{y}) = \lambda$, and put $j^\xi_\beta(M,m,x) = J_M(B)$. Thus j^ξ_β is defined precisely in the stable case.

An iteration over M belongs to $\mathcal{K}^\xi_{\beta+1}(M)$ if it belongs to $\mathcal{K}^\xi_\beta(M)$ and, whenever a stage codes (m,x,y), the extension N by the preceding initial segment satisfies

$$j^\xi_\beta(N,m,x)\downarrow \implies y \neq j^\xi_\beta(N,m,x). \tag{3}$$

Take intersections at limit levels. At a limit of hierarchies retain the union of the restrictions on coding and use every earlier fixed class in the definition at level zero.

Lemma 2.13 (Boolean values and stabilization)**.** *At the start of hierarchy ξ, the recursion defines*

$$a_\beta(H,\dot{T}) = \|\dot{T} \in \mathcal{K}^\xi_\beta\|_{B_H}$$

for every named iteration H and quotient name $\dot{T}$, where B_H is the Boolean algebra of H. These values commute with the translations along the specified embeddings, also over products. For some ordinal θ_ξ, the families $\langle a_\beta(H,\dot{T}) : H,\dot{T}\rangle$ are constant for $\beta \geq \theta_\xi$.

Proof. For a complete embedding $j : B_H \to B_{H*\dot{T}}$, put

$$\exists_j c = \min\{d \in B_H : c \leq j(d)\}, \qquad \forall_j c = \neg\exists_j \neg c.$$

Here min is in the Boolean order. These are the values of $c/G_H \neq 0$ and $c/G_H = 1$, respectively. Below a condition use the relative Boolean operations.

At level zero, $a_0(H,\dot{T})$ is the meet of the values of the coding restrictions and membership in all earlier fixed classes. Given the family a_β, let $j_{\dot{T}}$ be the embedding for $H * \dot{T}$. The values for persistence and attainability at $(m,\dot{x})$ are

$$\|\dot{b} \in P^\xi_\beta\|_{B_H} = \bigwedge_{\dot{T}} (\neg a_\beta(H,\dot{T}) \vee \forall_{j_{\dot{T}}} \|A(\dot{b})\|_{H*\dot{T}}),$$

$$\|\lambda \text{ is attained}\|_{B_H} = \bigvee_{\dot{T},\dot{y}} (a_\beta(H,\dot{T}) \wedge \exists_{j_{\dot{T}}} \|\rho_{\leq(\xi,\beta)}(m,\dot{x},\dot{y}) = \lambda\|_{H*\dot{T}}).$$

In the second line, conjoin $\dot{y} \notin \dot{\mathbb{R}}^H$ inside the projected value to test for a new minimizer, and conjoin $\dot{y} = \dot{z}$ for a lifted H-name $\dot{z}$ to test $\dot{z} \in B$ at the minimum. Leastness and the name for J_H give the value g_i of (3) in the algebra of the initial segment preceding stage i; put $g_i = 1$ if the almost disjoint factor is trivial. For the embeddings j_i from these initial segments into $H * \dot{T}$, the successor and limit clauses are

$$a_{\beta+1}(H,\dot{T}) = a_\beta(H,\dot{T}) \wedge \bigwedge_i \forall_{j_i} g_i,$$

$$a_\delta(H,\dot{T}) = \bigwedge_{\beta<\delta} a_\beta(H,\dot{T}) \qquad (\delta \text{ limit}).$$

Thus the successor requirement holds on every branch.

For an earlier hierarchy and $H = A * P$, where A ends at the start of a later hierarchy, choose the names for $P, \dot{T}$ below a condition in G_A and translate them to the earlier base by the specified embeddings. Then

$$a_\beta(H, \dot{T}) \in G_A * G_P \quad \Longleftrightarrow \quad a_\beta(H, \dot{T})/G_A \in G_P.$$

Names forced equal below a condition give equal values there; if their evaluations agree, first strengthen to force equality. Induction in the displayed Boolean clauses gives compatibility under the translations preserving the restrictions, embeddings, and choice functions. In a product the earlier embeddings go through the final-iteration factor and are retained when appending a further iteration. At a limit of hierarchies take the meet of the earlier quotient values.

The families a_β decrease in a fixed set. Choose θ_ξ with $a_{\theta_\xi} = a_{\theta_\xi + 1}$ as families. The persistence sets, ranks, minima and choices then agree at these two levels, so the successor and limit clauses give $a_\beta = a_{\theta_\xi}$ for all $\beta \geq \theta_\xi$. Write $K_\xi(M) = \mathcal{K}^\xi_{\theta_\xi}(M)$. □

Lemma 2.14 (Products and preservation of ranks)**.** *If $Q, R \in \mathcal{K}^\xi_\alpha(M)$, then*

$$(Q \times R)/G_Q \in \mathcal{K}^\xi_\alpha(M[G_Q]), \qquad (Q \times R)/G_R \in \mathcal{K}^\xi_\alpha(M[G_R]).$$

For $N = M[G_R]$, write ρ^M, ρ^N for the partial ranks through (ξ, α). Then

$$\rho^N \restriction (\omega \times \mathbb{R}^2)^M = \rho^M, \qquad \Lambda^\xi_\alpha(N, m, x) = \Lambda^\xi_\alpha(M, m, x) \quad (m, x \in M).$$

If the common minimum exists, it is stable over M iff it is stable over N. In the stable case the sets of minimizers agree and $j^\xi_\alpha(N, m, x) = j^\xi_\alpha(M, m, x)$.

Proof. Induct lexicographically on (ξ, α), proving the product assertions first. The definition and Lemma 2.13 give closure under

$$T \mapsto T \restriction t, \qquad T \mapsto T_i, \qquad T \mapsto T/G_{T_i}.$$

A name given below d is lifted using trivial almost disjoint factors on $\neg d$. Evaluating each requirement immediately before its coding stage also gives

$$T \in \mathcal{K}^\xi_\alpha(S), \qquad T \Vdash \dot{U} \in \mathcal{K}^\xi_\alpha(S[G_T]) \\ \Longrightarrow \quad T * \dot{U} \in \mathcal{K}^\xi_\alpha(S).$$

In $Q \times R$, each coding stage retains its factor's real, so the coding restrictions hold. Membership in earlier fixed classes follows from the induction on ξ, through the specified embeddings. For $\beta < \alpha$, let S be the extension preceding a stage of one factor and $S' = S[G_U]$ the extension obtained by adding the other factor. By induction,

$$U \in \mathcal{K}^\xi_\beta(S), \qquad j^\xi_\beta(S', m, x) = j^\xi_\beta(S, m, x),$$

where equality includes definedness. Thus (3) transfers from S to S', proving the product assertions at α.

Fix $N = M[G_R]$ and $b \in (\omega \times \mathbb{R}^2)^M$. We have

$$b \notin P^\xi_\alpha(N) \quad \iff \quad \exists T \in \mathcal{K}^\xi_\alpha(N)\ \exists t \in T\ t \Vdash^N_T \neg A(b).$$

Given such T, t, choose their names and decide the length below $r \in G_R$. Lift the iteration as above. Then

$$R * \dot{T}^\uparrow \in \mathcal{K}^\xi_\alpha(M), \qquad (r, \dot{t}) \Vdash \neg A(b),$$

so $b \notin P^\xi_\alpha(M)$. Conversely, if $Q \in \mathcal{K}^\xi_\alpha(M)$ and $q \Vdash^M_Q \neg A(b)$, the witness persists in $Q \times R$. Hence, identifying q with its quotient condition,

$$q \Vdash^N_{(Q\times R)/G_R} \neg A(b).$$

Consequently

$$P^\xi_\alpha(N) \cap M = P^\xi_\alpha(M).$$

Applying this at the lower levels and earlier hierarchies yields $\rho^N \restriction (\omega \times \mathbb{R}^2)^M = \rho^M$.

Composition $R * \dot{T}^\uparrow$ and the product $T \times R$ give, respectively, the two inclusions in

$$\Lambda^\xi_\alpha(N, m, x) = \Lambda^\xi_\alpha(M, m, x).$$

For the product direction, the quotient over $M[G_T]$ is allowable at α, so the witnessing rank is preserved. For new witnesses, generic intersection gives

$$M[G_T] \cap M[G_R] = M,$$
$$y \in M[G_T] \setminus M \quad \Longrightarrow \quad y \notin N.$$

Conversely, $y \notin N \Rightarrow y \notin M$ in a composed extension. Thus the common minimum is stable over M iff it is stable over N.

In the stable case, put $\lambda = \min \Lambda^\xi_\alpha(M, m, x)$ and let B_M, B_N be the minimizer sets. Composition gives $B_N \subseteq B_M$, and the product gives $B_M \subseteq B_N$. Rank preservation and the trivial forcing give

$$B_S = \{y \in \mathbb{R}^S : \rho^S(m, x, y) = \lambda\} \qquad (S = M, N).$$

For $B = B_M = B_N$, coherence therefore yields

$$j^\xi_\alpha(N, m, x) = J_N(B) = J_M(B) = j^\xi_\alpha(M, m, x).$$

□

In particular, whenever (3) from an earlier level or hierarchy forbids coding a triple over M, the minimum remains stable and the same triple is forbidden by the corresponding requirement at the current fixed level.

Indeed, a minimizer already in M witnesses λ using the trivial forcing in the smaller class. Any smaller rank attainable in the smaller class would already have been visible at the earlier level, since newly assigned ranks are later in the lexicographic order. Any new real of rank λ obtainable in the smaller class would also have been obtainable in the earlier class. Thus the set of minimizers and its choice agree. If the current minimum can be attained by a new real, no earlier requirement of this kind is defined at (m, x) over M.

Lemma 2.15 (Products after imposing new restrictions on coding)**.** *Let M be an extension by a bounded initial segment of the final iteration during hierarchy η, and let $R \in K_\eta(M)$ be a further bounded segment of the final iteration. Suppose*

$$(m, x, y) \text{ forbidden during } R \quad \Longrightarrow \quad y \notin M.$$

Let ξ be the current hierarchy in $M[G_R]$. For every $Q \in K_\eta(M)$, the quotient $(Q \times R)/G_R$ belongs to all the intervening fixed classes and to $K_\xi(M[G_R])$. For each initial segment Q_i, put

$$S = M[G_{Q_i}], \qquad N = M[G_{Q_i}, G_R].$$

Writing ρ^S, ρ^N for the partial ranks through (η, θ_η) and (ξ, θ_ξ), respectively, we have

$$\rho^N \restriction (\omega \times \mathbb{R}^2)^S = \rho^S.$$

For $(m, x) \in S$, if the minimum over S in $K_\eta(S)$ exists, then

$$\min \Lambda^\xi_{\theta_\xi}(N, m, x) = \min \Lambda^\eta_{\theta_\eta}(S, m, x).$$

The two minima have the same stability. In the stable case the sets of minimizers and the choices agree.

Proof. Fix the product presentation and induct on the later hierarchies and their levels, simultaneously for all $Q \in K_\eta(M)$ and all Q_i. Earlier Boolean values are evaluated through the final-iteration factor in any bounded extension containing the start of their hierarchy, also after a restart. Lemmas 2.12 and 2.13 retain the embeddings and choice functions under $Q_i \mapsto Q_i * \dot{T}$.

At level zero, the earlier fixed classes follow from Lemma 2.14 and induction on the hierarchies. A newly forbidden (m, x, y) cannot be coded by Q: at its preceding stage Q_i we would have

$$y \in M[G_{Q_i}] \cap M[G_R] = M, \qquad y \notin M.$$

At a limit of hierarchies intersect the earlier fixed classes.

Suppose the product assertion holds at level β of hierarchy ξ. Put

$$S = M[G_{Q_i}], \qquad N = M[G_{Q_i}, G_R], \qquad \rho^N_\beta = \rho^N_{\leq(\xi,\beta)}.$$

For $b \in (\omega \times \mathbb{R}^2)^S$,

$$\rho^S(b) \uparrow \implies \exists T \in K_\eta(S)\ \exists t \in T\ t \Vdash_T^S \neg A(b).$$

Replace T by $T \restriction t$, choose its name and length below a condition in G_{Q_i}, and lift it using trivial almost disjoint factors on the complement. Then

$$Q_i * \dot{T}^\uparrow \in K_\eta(M).$$

Extend the specified presentation of $Q_i \times R$ to $(Q_i * \dot{T}^\uparrow) \times R$. By induction its appended quotient

$$U = ((Q_i * \dot{T}^\uparrow) \times R)/(G_{Q_i} \times G_R)$$

satisfies

$$U \in \mathcal{K}_\beta^\xi(N), \qquad U \Vdash_N \neg A(b), \qquad b \notin P_\beta^\xi(N).$$

Here the witness to $\neg A(b)$ persists in the product. Previously assigned ranks agree by Lemma 2.14, evaluating the earlier definitions in this same product. Together with the induction,

$$\rho_\beta^N \restriction (\omega \times \mathbb{R}^2)^S = \rho^S.$$

Suppose $\lambda = \min \Lambda_{\theta_\eta}^\eta(S, m, x)$ exists. Let $T \in K_\eta(S)$ witness λ, restricted below a witnessing condition. Appending T as above gives

$$\lambda \in \Lambda_\beta^\xi(N, m, x).$$

In the product over S, generic intersection gives

$$S[G_T] \cap N = S, \qquad y \in S[G_T] \setminus S \implies y \notin N.$$

Thus a new witness remains new. Conversely, put $V = (Q_i \times R)/G_{Q_i}$. By Lemma 2.14, $V \in K_\eta(S)$. For a witnessing $U \in \mathcal{K}_\beta^\xi(N)$, choose and lift its name as above. Then

$$V * \dot{U}^\uparrow \in K_\eta(S).$$

Since every rank $\mu < \lambda$ already lies through (η, θ_η), this composition gives

$$\mu < \lambda, \quad \mu \in \Lambda_\beta^\xi(N, m, x) \implies \mu \in \Lambda_{\theta_\eta}^\eta(S, m, x),$$

contrary to minimality. Hence

$$\min \Lambda_\beta^\xi(N, m, x) = \lambda.$$

If the old minimum is stable, composition excludes new minimizers over S; the old minimizers retain rank λ. For the corresponding sets B_S, B_N, therefore,

$$B_N \subseteq B_S \subseteq B_N, \qquad j_\beta^\xi(N, m, x) = J_N(B_N) = J_S(B_S) = j_{\theta_\eta}^\eta(S, m, x).$$

Together with the preservation of new witnesses, this proves equality of stability.

At a stage of Q coding (m, x, y), if the old minimum exists, its stability and choice transfer (3) from S to N. Otherwise, the trivial forcing and rank preservation give

$$\Lambda^{\eta}_{\theta_\eta}(S, m, x) = \varnothing \implies \rho^S(m, x, y) \uparrow \implies \rho^N_\beta(m, x, y) \uparrow .$$

Thus $j^\xi_\beta(N, m, x) \downarrow$ implies $y \neq j^\xi_\beta(N, m, x)$ in this case as well. This proves the successor step. At limit levels take all earlier requirements, yielding the assertions at the fixed level. □

Lemma 2.16 (First initial segment containing a minimizer)**.** *Suppose the current fixed class allows its minimum rank λ at (m, x) to be attained by a real outside M. There is a restriction of an allowable iteration that forces a real y of rank λ at (m, x) to appear in its final extension and in no extension by a proper initial segment.*

Proof. Choose $Q \in K_\xi(M)$, $q \in Q$, and $\dot y$ with

$$q \Vdash_Q \dot y \notin \mathbb{R}^M \ \wedge \ \rho_{\leq(\xi,\theta_\xi)}(m, x, \dot y) = \lambda.$$

Strengthen q to decide $\delta > 0$ and a Q_δ-name $\dot z$ such that

$$\begin{aligned} &q \Vdash_Q \delta = \min\{\gamma \leq \mathrm{lh}(Q) : \dot y \in \dot{\mathbb{R}}^{Q_\gamma}\},\\ &q \Vdash_Q \dot y = \dot z, \end{aligned}$$

identifying names with their lifts. Since Q/G_{Q_δ} is allowable, Lemma 2.14 implies that q forces the Q_δ-extension to satisfy

$$\dot z \notin \bigcup_{\gamma<\delta} \dot{\mathbb{R}}^{Q_\gamma}, \qquad \rho_{\leq(\xi,\theta_\xi)}(m, x, \dot z) = \lambda.$$

Take $p \leq q$ in the dense presentation and put $p_\delta = \pi_\delta(p)$. Since

$$s \leq p_\delta \quad \implies \quad s \not\perp_Q p,$$

p_δ forces both assertions over Q_δ. Thus $Q_\delta \restriction p_\delta$, with initial segments $Q_\gamma \restriction \pi_\gamma(p)$, $\gamma \leq \delta$, is as required, also when δ is a limit. □

3 The uniformizing extension

Theorem 3.1. *There is a coding iteration P of length ω_1 over W such that, for every W-generic $G \subseteq P$, each A_m is uniformized in $W[G]$ by*

$$U_m(x, y) \quad \iff \quad (x, y) \in A_m \ \wedge \ \neg\Phi(\langle x, y, m\rangle).$$

Proof. We use Lemmas 2.9–2.11 and the absoluteness facts of Subsection 2.6. Let P_i be the initial segment of length i, and put $G_i = G \cap P_i$. By Lemma 2.10,

$$\omega_1^{W[G_i]} = \omega_1, \qquad \mathbb{R}^{W[G_i]} = \{x_{i,j} : j < \omega_1\}, \qquad \mathbb{R}^{W[G]} = \bigcup_{i<\omega_1} \mathbb{R}^{W[G_i]},$$

where each enumeration is specified by a P_i-name. If ξ_i is the hierarchy in use at stage i, abbreviate

$$\mathcal{K}_i = K_{\xi_i}(W[G_i]), \qquad \rho_i = \rho^{W[G_i]}_{\leq(\xi_i,\theta_{\xi_i})},$$
$$\Lambda_i(m,x) = \Lambda^{\xi_i}_{\theta_{\xi_i}}(W[G_i],m,x), \quad j_i(m,x) = j^{\xi_i}_{\theta_{\xi_i}}(W[G_i],m,x).$$

Let F_ξ be the triples explicitly forbidden in hierarchy ξ. List the following instructions cofinally often, using (i,j) for $x_{i,j}$; skip references to initial segments not yet constructed. Define $y_*(m,x)$ at most once.

1. For $b = (m,x,y) \in W[G_i]$,

   $$\rho_i(b)\!\uparrow \implies \exists Q \in \mathcal{K}_i\ \exists q \in Q\ q \Vdash^{W[G_i]}_Q \neg A(b).$$

   Append $Q \restriction q$.

2. If $\mathcal{K}_i$ permits coding (m,x,y), append a coding stage for $\langle x,y,m\rangle$.

3. Suppose $y_*(m,x)\!\uparrow$ and $\Lambda_i(m,x) \neq \varnothing$. Put $\lambda = \min \Lambda_i(m,x)$. If λ is stable, set

   $$y_*(m,x) = j_i(m,x) = J_{W[G_i]}(B),$$

   where B is the set of minimizers. Otherwise, Lemma 2.16 supplies $Q \in \mathcal{K}_i$, $\delta = \mathrm{lh}(Q) > 0$, and $\dot y$ such that

   $$Q \Vdash \dot y \notin \bigcup_{\alpha<\delta} \dot{\mathbb{R}}^{Q_\alpha} \quad \wedge \quad \rho_{\leq(\xi_i,\theta_{\xi_i})}(m,x,\dot y) = \lambda.$$

   Append Q, put $y_*(m,x) = \dot y^H$ for its generic H, and restart with

   $$F_{\xi_i+1} = F_{\xi_i} \cup \{(m,x,y_*(m,x))\}.$$

   The restriction applies before the next factor.

Implement the instructions by Lemma 2.13. On a maximal antichain $\mathcal{A} \subseteq P_i$, choose $\dot Q_a$ and $\delta_a < \omega_1$ with $a \Vdash \mathrm{lh}(\dot Q_a) = \delta_a$. At the k-th stage within a tail, mix its k-th factor names; at limits take the prescribed limits of initial segments. Thus the length of an instruction may depend on the generic. Between instructions extend the earlier choice functions coherently to all new nonempty sets of reals. At a limit restart ζ, use

$$F_\zeta = \bigcup_{\xi<\zeta} F_\xi, \qquad \mathcal{K}^\zeta_0(M) = \bigcap_{\xi<\zeta} K_\xi(M).$$

Each restart acts before the next factor; earlier hierarchies continue to determine previously assigned ranks. If no tail is appended, insert a stage with trivial almost disjoint factor after the choice or restart. For the starting stages of the instructions, therefore,

$$\gamma_0 = 0, \qquad \gamma_{\nu+1} = \gamma_\nu + \ell_\nu \quad (0 < \ell_\nu < \omega_1), \qquad \gamma_\delta = \sup_{\nu<\delta} \gamma_\nu \quad (\delta \text{ limit}),$$
$$\nu \le \gamma_\nu < \omega_1, \qquad \sup_{\nu<\omega_1} \gamma_\nu = \omega_1.$$

The last line follows from regularity of ω_1. Together with localization of reals, it ensures that each instruction is considered after its parameters have appeared.

Verification. For $i < j < \omega_1$, the construction gives

$$P_j/G_i \in \mathcal{K}_i,$$
$$(m, x, y) \in F_{\xi_j} \setminus F_{\xi_i} \quad \Longrightarrow \quad y \notin W[G_i].$$

Thus Lemmas 2.14 and 2.15 yield

$$\rho_j \restriction (\omega \times \mathbb{R}^2)^{W[G_i]} = \rho_i.$$

For $b \in (\omega \times \mathbb{R}^2)^{W[G_i]}$, persistence, the two absoluteness facts and instruction (1) give

$$\begin{aligned} \rho_i(b)\downarrow &\Longrightarrow \forall j \in [i, \omega_1) \quad W[G_j] \models A(b) \\ &\Longrightarrow W[G] \models A(b), \end{aligned}$$

$$\begin{aligned} \rho_i(b)\uparrow &\overset{(1)}{\Longrightarrow} \exists j \in [i, \omega_1) \quad W[G_j] \models \neg A(b) \\ &\Longrightarrow W[G] \models \neg A(b). \end{aligned}$$

Hence

$$W[G] \models A(b) \quad \Longleftrightarrow \quad \rho_i(b)\downarrow .$$

Using the trivial forcing and preservation of the least attainable rank,

$$\begin{aligned} A^{W[G]}_{m,x} \neq \varnothing &\Longrightarrow \exists i, y \quad x, y \in W[G_i] \ \wedge \ \rho_i(m, x, y)\downarrow \\ &\Longrightarrow \exists i \quad \Lambda_i(m, x) \neq \varnothing \\ &\overset{(3)}{\Longrightarrow} y_*(m, x)\downarrow \ \wedge W[G] \models A(m, x, y_*(m, x)). \end{aligned}$$

Fix $A^{W[G]}_{m,x} \neq \varnothing$, and let $y_* = y_*(m, x)$ be chosen at γ. Put

$$\lambda = \rho_\gamma(m, x, y_*), \qquad \mathcal{C} = \{u_i : i < \omega_1,\ i \text{ is a coding stage}\}.$$

If y_* was chosen at a stable minimum, preservation gives, for every $i < \omega_1$,

$$\begin{aligned} x, y_* \in W[G_i] &\Longrightarrow \rho_i(m, x, y_*)\downarrow \\ &\Longrightarrow \Lambda_i(m, x) \neq \varnothing \\ &\Longrightarrow j_i(m, x) = j_\gamma(m, x) = y_*. \end{aligned}$$

Therefore (3) gives $\langle x, y_*, m\rangle \notin \mathcal{C}$. Otherwise, instruction (3) gives, also for limit γ,

$$i < \gamma \implies y_* \notin W[G_i], \qquad i \geq \gamma \implies (m, x, y_*) \in F_{\xi_i},$$
$$\implies \quad \langle x, y_*, m\rangle \notin \mathcal{C}.$$

Let instruction (2) act on (m, x, y) at stage $i \geq \gamma$, where $y \neq y_*$. In the stable case, the observation after Lemma 2.14 gives, for every applicable guard,

$$j^\eta_\beta(W[G_i], m, x)\downarrow \quad \implies \quad j^\eta_\beta(W[G_i], m, x) = y_*.$$

Otherwise, if γ_ν starts the instruction choosing y_*, then $y_* \notin W[G_{\gamma_\nu}]$. By Lemma 2.15 over $W[G_{\gamma_\nu}]$ and the same observation,

$$\exists Q \in \mathcal{K}_i \ \exists q \in Q \ \exists \dot z$$
$$q \Vdash_Q^{W[G_i]} \big(\dot z \notin \mathbb{R}^{W[G_i]} \ \wedge \ \rho_{\leq(\xi_i,\theta_{\xi_i})}(m, x, \dot z) = \lambda\big),$$
$$\implies \quad j^\eta_\beta(W[G_i], m, x)\uparrow \quad \text{for every applicable guard.}$$

Since

$$\{z : (m, x, z) \in F_{\xi_i}\} \subseteq \{y_*\},$$

instruction (2) yields

$$y \neq y_* \quad \implies \quad u_i = \langle x, y, m\rangle \in \mathcal{C}.$$

By Lemma 2.11, in $W[G]$,

$$\Phi(\langle x, y, m\rangle) \quad \iff \quad \langle x, y, m\rangle \in \mathcal{C} \quad \iff \quad y \neq y_*.$$

Together with $U_m \subseteq A_m$, this gives

$$\{y : U_m(x, y)\} = \begin{cases} \{y_*(m, x)\}, & A_{m,x} \neq \varnothing, \\ \varnothing, & A_{m,x} = \varnothing. \end{cases}$$

Finally,

$$A_m \in \Pi^1_k, \quad \Phi \in \Sigma^1_k \quad \implies \quad U_m \in \Pi^1_k.$$

Include real parameters in the first coordinate of a universal relation. □

Remark. The arguments of Lemmas 2.12–2.16 and the construction above use only the iteration and coding properties in Lemmas 2.9–2.11 and the two absoluteness facts of Subsection 2.6. Thus the same argument applies over another ground model once the corresponding properties are verified. We use this in Subsection 4.5.

Corollary 3.2. *There is a cardinal-preserving set-generic extension of L in which* CH *and the Π^1_3-uniformization property hold.*

Proof. First force with the branch product producing W, then with the iteration of Theorem 3.1. Cardinal preservation and CH follow from Theorem 2.7 and Lemma 2.10. Since Φ is Σ^1_3, the uniformizing graphs are Π^1_3. □

4 Forcing over the canonical inner models M_n

Fix for the remainder of this section a natural number n with $1 \leq n < \omega$, and assume that $M_n^{\#}$ exists. As always we let M_n be the resulting inner model one obtains when iterating the sharp out of the universe. Throughout this section, ω_1 denotes $\omega_1^{M_n}$. Our goal is to prove the following theorem.

Theorem. *There is a set-generic extension of M_n in which the Π^1_{n+3}-uniformization property holds.*

The proof follows the construction over L from the preceding sections, but three ingredients require M_n-analogues: the recovery of the canonical lower part $M_n|\omega_1$, the definition of the coding predicate, and the absoluteness argument for witnesses to nonmembership in the sections. We begin with the inner-model-theoretic background needed for these analogues.

4.1 Inner-model-theoretic background and generic absoluteness

We use the notation and comparison conventions of Steel's analysis of mice with finitely many Woodin cardinals; see [20, 21]. An *ordinary premouse* means a premouse in the parameter-free premouse language. In particular, no real or other base set is included as a predicate.

Definition 4.1. *Let $\mathcal{M}$ be a premouse and let $\eta < \mathrm{Ord}^{\mathcal{M}}$. We say that $\mathcal{M}$ is* n-small above η *if, whenever E is an extender on the $\mathcal{M}$-sequence and*

$$\eta < \mathrm{crit}(E),$$

the initial segment $\mathcal{J}^{\mathcal{M}}_{\mathrm{crit}(E)}$ does not have n Woodin cardinals above η. We say that $\mathcal{M}$ is n-small *if it is n-small above* 0.

We use Steel's finite-level iterability predicates with their parity-dependent definitions [20, Definitions 1.4 and 1.6]. For a premouse $\mathcal{M}$ and a cutpoint η, write $\mathrm{It}_n(\mathcal{M}, \eta)$ for this predicate above η. We will use the cutpoint ω.

Steel's analysis gives the following complexity statement.

Fact 4.2. *For each fixed $1 \leq n < \omega$, the relation*

"*x codes a countable premouse which is Π_n-iterable above ω.*"

is Π^1_{n+1} in the real code x. Moreover, every countable realizable n-small premouse is Π_n-iterable at the indicated cutpoint.

We next specify the generic extensions in which the comparison lemma will be used. All cardinalities in the following definition are computed in M_n.

Definition 4.3. *A model N is a* small M_n-generic extension *if*

$$N = M_n[G]$$

for some cardinal-preserving forcing $\mathbb{P} \in M_n$ such that $M_n \models |\mathbb{P}| \leq \aleph_1$ and some M_n-generic filter $G \subseteq \mathbb{P}$. A small two-step extension *is an extension*

$$M_n[G] \subseteq M_n[G][H]$$

obtained from a two-step forcing $\mathbb{P} * \dot{\mathbb{Q}} \in M_n$ which is forced to be cardinal preserving and of size at most $\aleph_1$.

The preparatory branch forcing and every forcing iteration used later will be shown to have size $\aleph_1$ and to preserve cardinals. Hence all models which occur in the proof are small M_n-generic extensions in the sense of Definition 4.3.

The second background ingredient is the mouse-operator closure which yields projective generic absoluteness. Recall that a set is *self-wellordered*, abbreviated *swo*, if it codes a wellorder of itself. For a self-wellordered set X, let $M_k^{\#}(X)$ denote the canonical active X-mouse with k Woodin cardinals. Thus $M_0^{\#}(X)$ is the usual sharp over X.

Fact 4.4. *The model M_n is closed under the operator*

$$X \longmapsto M_{n-1}^{\#}(X)$$

below its least Woodin cardinal. In particular, for every self-wellordered $X \in H(\omega_2)^{M_n}$, the mouse $M_{n-1}^{\#}(X)$ exists and is normally ω_2-iterable in M_n.

This closure is preserved by the small forcing notions which occur in the construction.

Lemma 4.5. *Let $N = M_n[G]$ be a small M_n-generic extension. Then*

$$N \models \text{“}M_{n-1}^{\#}(X) \text{ exists and is normally } \omega_2\text{-iterable for every swo } X \in H(\omega_2)\text{.”}$$

Moreover, whenever X belongs to both M_n and N, the two models compute the same mouse $M_{n-1}^{\#}(X)$.

Proof. Apply the generic preservation theorem for mouse operators to $M_{n-1}^{\#}$. In the notation of [19, Lemma 3.4], take $\kappa = \omega_1$ and replace the index occurring there by $n-1$. Fact 4.4 supplies the hypothesis that $M_{n-1}^{\#}(X)$ exists for every self-wellordered $X \in H(\omega_2)^{M_n}$. The cited lemma gives existence, normal ω_2-iterability, and correctness of the operator in the forcing extension. □

We can now state the precise generic absoluteness principle used later. It is the two-step form which permits real parameters introduced by the first forcing.

Theorem 4.6. *Let*

$$M_n[G] \subseteq M_n[G][H]$$

be a small two-step extension. Then

$$M_n[G] \prec_{\Sigma^1_{n+2}} M_n[G][H].$$

Equivalently, for every real parameter $a \in M_n[G]$ and every Σ^1_{n+2} formula $\vartheta(v)$,

$$M_n[G] \models \vartheta(a) \quad \Longleftrightarrow \quad M_n[G][H] \models \vartheta(a).$$

The same equivalence holds for Π^1_{n+2} formulas.

Proof. By Lemma 4.5, the intermediate model $M_n[G]$ is closed under $X \mapsto M^{\#}_{n-1}(X)$ for self-wellordered $X \in H(\omega_2)^{M_n[G]}$. Apply the Martin–Solovay–Schindler generic absoluteness theorem in the form of [19, Lemma 3.7], with the mouse index $n-1$ and $\kappa = \omega_1$, to the second forcing. This yields Σ^1_{n+2}-elementarity between $M_n[G]$ and $M_n[G][H]$. The Π^1_{n+2} assertion follows by taking complements. $\square$

The point at which Theorem 4.6 enters the uniformization proof is recorded in the following consequence.

Corollary 4.7. *Let*

$$M_n[G] \subseteq M_n[G][H]$$

be a small two-step extension, let $a \in M_n[G]$ be a real, and let $\varphi(v)$ be a Σ^1_{n+3} formula. If

$$M_n[G] \models \varphi(a),$$

then

$$M_n[G][H] \models \varphi(a).$$

Consequently, if $A \subseteq (2^\omega)^2$ is Π^1_{n+3} and

$$M_n[G] \models (x,y) \notin A,$$

then $(x,y) \notin A$ remains true in $M_n[G][H]$.

Proof. Write

$$\varphi(a) \equiv \exists z\, \psi(z,a),$$

where ψ is Π^1_{n+2}. Choose a witness $z \in M_n[G]$. By Theorem 4.6, the statement $\psi(z,a)$ remains true in $M_n[G][H]$, so the same witness establishes $\varphi(a)$ there. The final assertion is the special case in which $\varphi(x,y)$ defines the complement of A. $\square$

4.2 Recognition of the lower part

Definition 4.8. *Let $\mathcal{I}_n$ be the set of real codes for well-founded ordinary premice P such that*

$$P \text{ is } \omega\text{-sound}, \qquad \rho_\omega(P) = \omega,$$
$$P \text{ is } n\text{-small}, \qquad \mathrm{It}_n(P, \omega).$$

Here ω is a cutpoint, and we use Steel's iterability predicate from [20, Definitions 1.4 and 1.6]. We also write $P \in \mathcal{I}_n$ when a real code for P belongs to this set. For an ambient model N, let $\mathcal{C}_n^N$ be the class of transitive premouse structures whose codes in N satisfy the definition in N.

The definition of $\mathcal{I}_n$ is parameter-free Π^1_{n+1} and is invariant under a change of the enumeration of the coded premouse. Indeed the finite iterability predicate is Π_n over HC by [20, Lemmas 1.5 and 1.7], giving the bound in real codes in Fact 4.2. The remaining checks use the satisfaction relations of the coded structures and the finite-degree fine-structural definitions. They do not increase the bound, and well-foundedness is Π^1_1.

Lemma 4.9. *Every member of $\mathcal{C}_n^{M_n}$ is an initial segment of M_n of height below ω_1, and their heights are cofinal in ω_1. If N is a small M_n-generic extension, then*

$$\mathcal{C}_n^N = \mathcal{C}_n^{M_n}, \qquad \bigcup \mathcal{C}_n^N = M_n|\omega_1.$$

Proof. First work in M_n. Its reals are captured by the realizable, sound, projectum-ω segments in Steel's construction [20, Theorem 2.3]; see also [21, Corollaries 3.12–3.13]. These segments are countable in M_n and have unbounded heights in ω_1, since a bound would put every real of M_n in a single countable level. In the background for this construction M_n is Σ_n^{HC}-correct by [20, Corollary 4.4]. Thus the iterability predicate, which is Π_n over HC, agrees between M_n and the background on the relevant premice. The realizable segments satisfy the predicate there by [20, Lemma 2.2(2)] and hence belong to $\mathcal{C}_n^{M_n}$.

If $P \in \mathcal{C}_n^{M_n}$, choose a realizable segment Q of strictly greater height from this cofinal system. Both are ω-mice in the sense of [20, Definition 1.3]. They agree below the common cutpoint ω. By [20, Lemma 2.2(3)] they are comparable by initial segment, so $P \trianglelefteq Q \triangleleft M_n$. This proves ground recognition and cofinality.

Now let N be a small extension. For premouse codes p, q, let $\mathrm{Comp}(p, q)$ assert that either structure is isomorphic to an initial segment of the other. This has an analytic presentation: an initial-segment cut and an isomorphism are coded by a real. If $q \in M_n$ codes a recognized segment, then

$$\exists p \, (p \in \mathcal{I}_n \ \wedge \ \neg \, \mathrm{Comp}(p, q))$$

is $\Sigma^1_{n+2}(q)$ and is false in M_n by ground recognition. It remains false in N by Theorem 4.6, using the trivial forcing as the first step.

If N recognizes a premouse P, its height is below ω_1 since ω_1 is preserved. Choose $Q \in \mathcal{C}_n^{M_n}$ above this height and fix an old code q for Q. The preceding argument gives $P \trianglelefteq Q$. Consequently P has an old real code, which satisfies the predicate in N by isomorphism invariance, and hence in M_n by the same absoluteness theorem. Conversely every old recognized code still passes the test by absoluteness. Thus the recognized structures agree, and their cofinal union is $M_n|\omega_1$. □

Canonical trees. Write $M = M_n$ and use $<_M$ for its canonical construction order. The order on any initial segment is its own construction order, and every predecessor of an element of that segment belongs to it [20, 21]. All the following choices are made in M.

Lemma 4.10. *There is a canonical $\diamondsuit_{\omega_1}$ sequence $\langle a_\lambda : \lambda < \omega_1\rangle$ over M whose bounded computations agree between initial segments containing them.*

Proof. Set $a_\lambda = \varnothing$ at zero and successors. At a nonzero limit $\lambda < \omega_1$, choose the $<_M$-least pair (X, C) such that $X \subseteq \lambda$, C is club in λ, and $a_\gamma \neq X \cap \gamma$ for all $\gamma \in C$. Put $a_\lambda = X$. There is a witness at every such λ. Choose a strictly increasing cofinal sequence $\langle \gamma_j : j < \omega\rangle$ in λ with $\gamma_j > j$. For $C = \{\gamma_j : j < \omega\}$, take $X \subseteq \omega$ with $j \in X$ iff $j \notin a_{\gamma_j}$. The witness test uses only the sequence already constructed and the indicated ordinal and sets. Thus any initial segment containing the recursion and the chosen witness sees the same least pair. In a bounded computation the least-pair clause requires a witness.

The verification that this recursion gives diamond is Jensen's condensation argument in M [21, Theorems 5.1 and 5.4(1)]. For the least counterexample (X, C) at ω_1, use an ω-sound initial segment Q containing X, C and the sequence, with $\rho_\omega(Q) = \omega_1^Q = \omega_1$. The projectum calculation for this choice is the one in [21, Sections 2 and 5]. New subsets of ω_1 appear cofinally below $M|\omega_1^+$, giving sound levels projecting to at most ω_1. A sound level containing ω_1 cannot project below ω_1, since the hull of a smaller projectum and the standard parameters would be countable in M. Take a countable elementary hull of Q containing these data and the standard parameters up to a degree witnessing $\rho_\omega(Q) = \omega_1$. If δ is its intersection with ω_1, the inverse collapse has critical point and collapsed projectum both equal to δ. The condensation alternative involving an extender of length δ is excluded here. Its critical point $\nu < \delta$ is countable in Q. A surjection $s : \omega \to \nu$ in Q would have the same values in the degree-zero ultrapower, since every value is below the critical point, contrary to its range being $j(\nu)$. The collapse is therefore an initial segment of M.

The collapsed pair $(X \cap \delta, C \cap \delta)$ is the least counterexample to the sequence below δ in that segment. The bounded witness test and the initiality of the

construction order make it least in M as well. Since $\delta \in C$, the identity $a_\delta = X \cap \delta$ contradicts the counterexample. $\square$

Use this diamond in the simultaneous construction of Lemma 2.5, with the $<_M$-least choice at every step. It gives independent normal splitting Suslin trees $\vec{S}^n = \langle S^n_\xi : \xi < \omega_1 \rangle$ in M. In particular, the finite product argument there seals each guessed maximal antichain in the appropriate product. At each level use the $<_M$-least enumeration by an initial segment of ω to label its nodes. These labels will be used in the decoder.

4.3 The preparatory branch extension W_n

We now pass to the ground model over which the coding iteration will be performed. Let

$$\vec{S}^n = \langle S^n_\xi \mid \xi < \omega_1 \rangle$$

be the independent sequence just constructed. All products in this subsection are computed in M_n.

Definition 4.11. *For $A \subseteq \omega_1$ with $A \in M_n$, put*

$$\mathrm{Br}^n_A := \prod_{\xi \in A}^{\mathrm{fs}} S^n_\xi.$$

We write

$$\mathrm{Br}_n := \mathrm{Br}^n_{\omega_1} = \prod_{\xi < \omega_1}^{\mathrm{fs}} S^n_\xi.$$

Let $H \subseteq \mathrm{Br}_n$ be M_n-generic. For $\xi < \omega_1$, define

$$b_\xi := \bigcup \{ p(\xi) : p \in H \text{ and } \xi \in \mathrm{dom}(p) \}.$$

For $A \subseteq \omega_1$ in M_n, let

$$H_A := H \cap \mathrm{Br}^n_A \,.$$

Finally, put

$$W_n := M_n[H].$$

Thus b_ξ is the branch added through S^n_ξ by the ξ-th coordinate. As before, to *destroy* a tree S^n_ξ means to add a cofinal generic branch through it.

Lemma 4.12. *Let $A, B \in M_n$ satisfy*

$$A \subseteq B \subseteq \omega_1.$$

Then the following hold in M_n.

1. *Br^n_A is c.c.c.*

2. Br^n_A *is a complete subforcing of* Br^n_B*, and*

$$\mathrm{Br}^n_B \cong \mathrm{Br}^n_A \times \mathrm{Br}^n_{B\setminus A}\,.$$

3. *If* $\xi \notin A$*, then*

$$\mathrm{Br}^n_A \Vdash \text{“}S^n_\xi \text{ is a Suslin tree.”}$$

Proof. The sequence $\vec{S}^n$ is independent in M_n, so every nonempty finite product of its members is a Suslin tree. Apply Lemma 2.2 inside M_n. The second clause is the coordinatewise factorization of the finite-support product. For the last clause, note that

$$\mathrm{Br}^n_A \times S^n_\xi \cong \mathrm{Br}^n_{A\cup\{\xi\}},$$

which is c.c.c. by the first clause. The product characterization of preservation of a Suslin tree therefore gives the conclusion. □

Theorem 4.13. *The model* $W_n = M_n[H]$ *has the following properties.*

1. *The forcing* Br_n *is c.c.c. and has size* $\aleph_1$ *in* M_n*. Consequently,* W_n *preserves every cardinal and cofinality of* M_n*, and*

$$W_n \models \mathsf{CH}.$$

2. *Every* S^n_ξ *is destroyed in* W_n*, as witnessed by the branch* b_ξ*.*

3. *If* $A \subseteq \omega_1$ *belongs to* M_n*, then* H_A *is* Br^n_A*-generic over* M_n*, and, for every* $\xi < \omega_1$*,*

$$M_n[H_A] \models \text{“}S^n_\xi \text{ has a cofinal branch”} \quad\Longleftrightarrow\quad \xi \in A.$$

4. *Every real in* W_n *belongs to* $M_n[H_A]$ *for some countable set* $A \subseteq \omega_1$ *with* $A \in M_n$*.*

5. *More generally, every member of* $H(\omega_1)^{W_n}$ *belongs to* $M_n[H_A]$ *for some countable set* $A \subseteq \omega_1$ *with* $A \in M_n$*.*

Proof. This is the argument of Theorem 2.7, carried out inside M_n, with $\vec{S}$, Br, and L replaced by $\vec{S}^n$, Br_n, and M_n, respectively. □

The last two clauses are the branch-coordinate localization needed by the coding argument. We record a form in which a prescribed countable set of coordinates may simultaneously be retained.

Corollary 4.14. *Let $a \in H(\omega_1)^{W_n}$, and let $C \in [\omega_1]^\omega \cap M_n$. Then there is $A \in [\omega_1]^\omega \cap M_n$ such that*

$$C \subseteq A, \qquad a \in M_n[H_A],$$

and

$$M_n[H_A] \models \text{“}S^n_\xi \text{ is a Suslin tree”} \qquad (\xi \in \omega_1 \setminus A).$$

In particular, no cofinal branch through any S^n_ξ with $\xi \notin A$ belongs to $M_n[H_A]$.

Proof. By Theorem 4.13(5), choose a countable branch support $A_0 \in M_n$ for a, and put

$$A = A_0 \cup C.$$

Then A is countable and belongs to M_n. Since $\mathrm{Br}^n_{A_0}$ is a complete subforcing of Br^n_A, we have $a \in M_n[H_A]$. The assertion about the unused trees follows from Lemma 4.12(3). □

4.4 The coding over M_n

Write $m^n_\infty = M_n|\omega_1$ and use its fixed ordinal code, with all premouse predicates, in $L[m^n_\infty, r]$. This code is in M_n, and its transitive collapse recovers the lower part.

The almost disjoint family. Fix a recursive injection $\#\colon 2^{<\omega} \to \omega$. In M define, for $\xi < \omega_1$,

$$x_\xi = \min_{<_M}\bigl(2^\omega \setminus \{x_\zeta : \zeta < \xi\}\bigr), \qquad d_\xi = \{\#(x_\xi \restriction k) : k < \omega\}.$$

Here $<_M$ is the canonical construction order on M. Each step is defined because ξ is countable and $M \models \mathsf{CH}$. The set d_ξ is infinite, and if $\zeta < \xi$ and k is the first position at which x_ζ and x_ξ differ, then

$$d_\zeta \cap d_\xi = \{\#(x_\xi \restriction j) : j \leq k\}.$$

Thus $D^n = \langle d_\xi : \xi < \omega_1\rangle$ is almost disjoint. Write $D = D^n$ throughout this subsection. Its initial segments are computed in initial segments of M below ω_1, as shown below.

For $Y \subseteq \omega_1$, the conditions of $\mathbb{A}^n(Y)$ are pairs $(s, F) \in [\omega]^{<\omega} \times [Y]^{<\omega}$, ordered by

$$(t, G) \leq (s, F) \quad\Longleftrightarrow\quad s \subseteq t, \quad F \subseteq G, \quad (t \setminus s) \cap \bigcup_{\xi \in F} d_\xi = \varnothing.$$

Conditions with a common stem are centered. For the union r of the generic stems, $\xi \in Y$ implies $r \cap d_\xi$ is finite by the density of conditions with $\xi \in F$. For $\xi \notin Y$ one can add arbitrarily many points of d_ξ to the stem, since each intersection $d_\xi \cap d_\zeta$ for $\zeta \in F$ is finite. Thus $Y = \{\xi < \omega_1 : |r \cap d_\xi| < \aleph_0\}$.

The coding sets. Fix in M a bijection $\varrho_n\colon ({}^{<\omega_1}2)^M \to \omega_1$, using CH in M. Let $\mathbb{C}_{M_n} = ({}^{<\omega_1}2)^M$, ordered by end-extension. This is the dense presentation by initial segments of $\mathrm{Add}(\omega_1,1)^M$. For a $\mathbb{C}_{M_n}$-generic g over the current model set

$$h(g) = \{\varrho_n(g \restriction \eta) : \eta < \omega_1\},$$
$$\mathrm{Sel}(h,u) = \{\tau_u(\gamma,k) : \gamma \in h,\ k<\omega\}, \qquad \tau_u(\gamma,k) = \omega\gamma + 2k + u(k).$$

We call $h(g)$ the *coding set* of g. Its cardinality is ω_1. For distinct g and g' the intersection $h(g)\cap h(g')$ is bounded in ω_1, since it consists of the images of their common initial segments. The decoding uses the set h itself and not ϱ_n.

To code $u \in 2^\omega$, let g be the generic real at a fresh Cohen coordinate and put $h = h(g)$. The forcing at the second step is $\mathbb{A}^n(Y_n(u,g))$, where $Y_n(u,g) \subseteq \omega_1$ is defined below. We give the localization explicitly. It codes three objects: u, h, and the branches with indices in $\mathrm{Sel}(h,u)$.

Codes and restriction to countable heights. Fix a finite fragment of ZFC^- containing the instances of separation, collection, replacement, set-structure satisfaction and wellorder collapse used below, and add the axiom that ω_1 exists. A *decoding model* is a countable transitive model of this theory. Let $\langle\cdot,\cdot\rangle_{\mathrm{ord}}$ be Gödel's pairing function on ordinals, which enumerates pairs by increasing maximum and lexicographically for a fixed maximum; for triples iterate it. If N is a decoding model and $\alpha = \omega_1^N$, then pairing and its inverse commute with restriction to α. Indeed, the part of the enumeration through any pair of ordinals below α is countable in N. Conversely the rank of a pair in the enumeration is at least the maximum of its entries. The same restriction property holds for $\zeta \mapsto j\cdot\zeta + i$ with $i<j<\omega$. Also $\tau_u(\gamma,k) < \alpha$ for $\gamma<\alpha$ and $k<\omega$, since $\alpha = \omega_1^N$.

Present the nodes of each S^n_ξ by pairs (λ, i), where λ is the level and i is the index in the $<_M$-least bijection from an initial segment of ω onto that level. The bijection is chosen as soon as the level has been constructed. We use these relabelled trees throughout, with the tree orders transported along the bijections. In the codes below a node is identified with the ordinal pairing of λ, i. Thus the restriction of the indexed tree to level $<\alpha$ is obtained by keeping exactly the node indices below α. We regard each b_ξ as the set of indices of the nodes of its downward closure.

Localization. For $\gamma \in h$ and $k<\omega$ use the index $\tau_u(\gamma,k)$ and put

$$B = \{\langle\gamma,k,\zeta\rangle_{\mathrm{ord}} : \gamma\in h,\ k<\omega,\ \zeta \in b_{\tau_u(\gamma,k)}\},$$
$$X = \{3\cdot k : k<\omega,\ u(k)=1\} \cup \{3\cdot\gamma+1 : \gamma\in h\} \cup \{3\cdot z+2 : z\in B\}.$$

The odd part of $Y_n(u,g)$ codes X. For the even part, define a continuous increasing sequence in $\mathrm{acc}(h)$ by

$$c_0 = \min\{\zeta \in \mathrm{acc}(h) : \omega < \zeta\},$$
$$c_{\nu+1} = \min\{\zeta \in \mathrm{acc}(h) : c_\nu + \omega < \zeta\}, \quad c_\lambda = \sup_{\nu<\lambda} c_\nu \quad (\lambda \text{ a nonzero limit}).$$

Fix in M a family $\langle w_\eta : \omega \le \eta < \omega_1\rangle$ with w_η coding a wellorder of ω of type η. Define

$$E = w_{c_0} \cup \{c_\nu + k : \nu < \omega_1,\ k \in w_{c_{\nu+1}}\},$$
$$Y_n(u,g) = \{2\cdot\zeta : \zeta \in E\} \cup \{2\cdot\zeta + 1 : \zeta \in X\}.$$

Here the wellorder codes use the fixed recursive coding of relations on ω. As in the construction over L, the even part localizes the code. The odd part gives the branch data. Apart from data in M, it uses only g, u and the selected branches.

Suppose N is a decoding model, with $\alpha = \omega_1^N$, and $Y_n(u,g) \cap \alpha \in N$. Then α belongs to $C = \{c_\nu : \nu < \omega_1\}$, which is club because the sequence is continuous, strictly increasing and has length ω_1. If $\alpha < c_0$, N reads w_{c_0} from the initial ω positions of the even part. If $c_\nu < \alpha < c_{\nu+1}$, then $c_\nu + \omega < \alpha$ by countability of c_ν in N. Using c_ν as an ordinal parameter, N reads $w_{c_{\nu+1}}$ from that block. In either case the relation read in N is well-founded in the ambient model, hence in N. Its collapse in N is an isomorphism onto an ordinal below α, since its domain is ω. The same map is an isomorphism in the ambient model, contradicting the prescribed order type. These cases exhaust the complement of C, since C is club. It follows that $h \cap \alpha$ is unbounded in α. The odd part yields $u, h\cap\alpha$ and, for $\gamma \in h\cap\alpha$, $k < \omega$, the branch $b_{\tau_u(\gamma,k)}$ restricted to levels below α. The restriction has one node at every such level, by the choice of node indices.

Correct local sequences. The set $\mathcal{I}_n$ is given by Definition 4.8. By Lemma 4.9, its recognized structures are unchanged in the small extensions used here. For each $\alpha < \omega_1$ the intended local sequences are

$$D^\alpha = D \restriction \alpha, \qquad \vec{S}^{n,\alpha} = \langle S^n_\xi \restriction \alpha : \xi < \alpha\rangle,$$

where the latter restriction is by levels. A triple $(m, D', \vec{S}')$ consists of a nonempty well-founded premouse of height $\alpha = \mathrm{Ord}(m)$ which is the union of its proper initial segments, together with sequences of the indicated shape. The triple is *correct* if for every $\tau < \alpha$ there is $\beta \in (\tau, \alpha)$ such that $m|\beta \in \mathcal{I}_n$ and a computation in $m|\beta$ agrees with $D', \vec{S}'$ below τ. This means agreement on the entries of D' of index $< \tau$ and on the trees of index $< \tau$ restricted to levels $< \tau$, including node indices and orders. A *computation* is an element of $m|\beta$ recording an initial part of the constructions of D and $\vec{S}^n$.

We show that correct triples agree with M, and that they exist at club many heights. The canonical order of M restricts to the order of each initial

segment, and every $<_M$-predecessor of an element of an initial segment belongs to it. A computation of D records the sequence of the x_ξ's and the sets of their finite initial segments. The checks at each entry are

$$x_\xi \in 2^\omega, \qquad x_\xi \notin \{x_\zeta : \zeta < \xi\}, \qquad d_\xi = \{\#(x_\xi \restriction j) : j < \omega\},$$
$$(\forall y \in 2^\omega)(y <_M x_\xi \implies \exists \zeta < \xi \ (y = x_\zeta)).$$

The quantifier over y ranges over the initial segment containing the computation. All earlier possible choices belong to that segment, so induction on ξ shows agreement with the construction in M.

For the diamond sequence the computation records the witnesses to the least-pair recursion of Lemma 4.10. A bounded computation at a nonzero limit stage includes the pair, with leastness checked in its segment. The test is absolute once the earlier values are fixed, so the same induction applies.

For the construction of the trees, the computation records the tree orders below the current level, the antichains coded by the diamond guesses, the families of branches chosen at limit levels, and the level bijections. If the guess at a limit λ codes a maximal antichain A in the product of the trees with indices in a finite set e, restricted to levels below λ, the recorded family $\langle b_{\xi,j} : \xi \in e, \ j < \omega \rangle$ must seal A:

$$\forall f \in {}^e\omega \ \exists t \in A \ \forall \xi \in e \ (t(\xi) \in b_{\xi,f(\xi)}).$$

Here maximality refers to this countable product. In addition every node below level λ of each tree must lie on one of the recorded branches. Such families exist by the countable product argument of Lemma 2.5. The conditions on the choices only refer to the data constructed so far. Hence any two initial segments of M containing these data and a witness make the same least choice, since all earlier possible choices lie in the smaller segment. An initial segment of the sequences together with its computation is hereditarily countable in M, hence belongs to $M|\omega_1 = H^M_{\omega_1}$ by the fine structure of the hierarchy [21, Section 2]. By Lemma 4.9 it belongs to a recognized segment below ω_1. Consequently we may fix in M a function f on ω_1 such that $f(\tau) > \tau$ is the height of an initial segment in $\mathcal{I}_n$ containing the computations up to τ.

Correctness is a parameter-free Π^1_{n+1} property: well-foundedness of the codes is Π^1_1, the conditions on computations inside the coded structures are arithmetic, and the quantifiers over initial segments and computations range over the countable codes. The definition of $\mathcal{I}_n$ and the agreement of computations give

$$(m, D', \vec{S}') = (M|\alpha, D^\alpha, \vec{S}^{n,\alpha})$$

for every correct triple. To see that correct triples exist at club many heights, fix a sufficiently large H_Θ in the current small extension. Take

countable $A \prec H_\Theta$ containing the parameters under consideration, m^n_∞, ω_1, the sequences $D, \vec{S}^n$, f, the map $\lambda \mapsto M|\lambda$, and ground enumerations of each $M|\lambda$, $0 < \lambda < \omega_1$. For $\alpha = A \cap \omega_1$, the enumerations and the level map give $A \cap m^n_\infty = M|\alpha$. The collapse N of A has $\omega_1^N = \alpha$, and sends $D, \vec{S}^n$ to $D^\alpha, \vec{S}^{n,\alpha}$. For $\xi < \alpha$, the node indices of S^n_ξ that are in A are exactly those below α, namely the indices of nodes at levels below α. These indices and the order between them are preserved by the collapse. Closure under f provides the required computations below α. The heights of a continuous chain of such submodels form a club.

Lemma 4.15 (Decoding)**.** *There are parameter-free formulas*

$$\Psi_n(r,u) \in \Pi^1_{n+2}, \qquad \Phi_n(u) \iff \exists r\, \Psi_n(r,u), \qquad \Phi_n \in \Sigma^1_{n+3},$$

with the following properties.

1. *If r is the real added at a coding stage for u, then $\Psi_n(r,u)$ holds in the resulting extension and in every further small extension.*

2. *If $\Psi_n(r,u)$ holds in a small extension, then $u \in L[m^n_\infty, r]$, and there is an unbounded $h \subseteq \omega_1$ in $L[m^n_\infty, r]$ such that $L[m^n_\infty, r]$ contains a cofinal branch through S^n_ξ for every $\xi \in \mathrm{Sel}(h,u)$.*

Proof. Let $\mathrm{Dec}_n(c; r, u)$ assert that c codes a decoding model N containing r, u and a triple $(m, D', \vec{S}')$ correct in the ambient model, with $\mathrm{Ord}(m) = \omega_1^N = \alpha$. In N put

$$Y^N_r = \{\xi < \alpha : |r \cap d'_\xi| < \omega\}.$$

Read (u', h', B') from its odd part and set $B'_{\gamma,k} = \{\zeta : \langle \gamma, k, \zeta\rangle_{\mathrm{ord}} \in B'\}$. Let $\mathrm{Br}_\alpha(b, S)$ mean that $b \subseteq S$ is linearly ordered by the tree order and meets every level below α exactly once. The local test is

$$\mathrm{Test}(c; r, u) \iff u' = u \wedge \sup h' = \alpha \wedge \forall \gamma \in h'\, \forall k < \omega\ \mathrm{Br}_\alpha(B'_{\gamma,k}, S'_{\tau_u(\gamma,k)}).$$

The code includes satisfaction relations checked arithmetically by the Tarski clauses. Correctness gives $\mathrm{Dec}_n \in \Pi^1_{n+1}$, and Test is arithmetic in the code. Hence

$$\Psi_n(r,u) \iff \forall c\, (\mathrm{Dec}_n(c; r, u) \implies \mathrm{Test}(c; r, u))$$

is parameter-free Π^1_{n+2}.

For r added by $\mathbb{A}^n(Y_n(u,g))$, possibly restricted below a condition, put $Y_r = \{\xi < \omega_1 : |r \cap d_\xi| < \omega\}$. Correctness and localization give, for every decoding model for (r,u),

$$Y_r = Y_n(u,g), \qquad Y^N_r = Y_r \cap \alpha,$$
$$(u', h', B') = (u, h \cap \alpha, B \cap \alpha), \qquad \alpha \in \mathrm{acc}(h).$$

For each $\gamma \in h \cap \alpha$, $k < \omega$, the section $B'_{\gamma,k}$ is $b_{\tau_u(\gamma,k)} \restriction \alpha$. Thus the test holds. Theorem 4.6 gives persistence.

Conversely, suppose $\Psi_n(r,u)$. In $L[m_\infty^n, r]$, recover $D, \vec{S}^n$ by taking the union of the computations in m_∞^n. They agree by induction on the least choices, and their bounds are cofinal in ω_1. Read $(\hat{u}, \hat{h}, \hat{B})$ from Y_r. At every height α of a decoding model for (r,u), correctness and restriction of the ordinal coding give

$$Y_r^N = Y_r \cap \alpha, \qquad (u', h', B') = (\hat{u}, \hat{h} \cap \alpha, \hat{B} \cap \alpha).$$

Such heights contain a club, so the tests give $\hat{u} = u$ and $\sup \hat{h} = \omega_1$. For each $\gamma \in \hat{h}$, $k < \omega$, and $\xi = \tau_u(\gamma, k)$, those tests at heights $\alpha > \gamma, \xi$ give

$$\mathrm{Br}_\alpha(\hat{B}_{\gamma,k} \cap \alpha, S_\xi^n \restriction \alpha).$$

Any two nodes and any prescribed level are covered at such a height. Thus $\mathrm{Br}_{\omega_1}(\hat{B}_{\gamma,k}, S_\xi^n)$, with $\hat{B}_{\gamma,k} \in L[m_\infty^n, r]$. □

Coding iterations and exactness

For a set $E \in M_n$ of at most $\aleph_1$ Cohen indices, put

$$C_E^n = \mathrm{Fn}_{<\omega_1}(E \times \omega_1, 2)^{M_n}.$$

Use the same sets E_i of Cohen indices as in the construction over L, with the reindexing maps now taken in M_n. For $\beta \leq \omega_1$, the forcing through the stages $i < \beta$ is

$$P_\beta^n = C_{E_{<\beta}}^n * \dot{R}_\beta^n, \qquad E_{<\beta} = \bigcup_{i<\beta} E_i, \qquad R_\beta^n = \operatorname*{fs}_{i<\beta}{}^{*} \dot{A}_i.$$

At stage i, names over P_i^n specify whether to code a real, the real u_i to be coded, the selected coordinate in E_i, and the ground-model isomorphism onto a restriction of $C_{E_i}^n$. If the stage codes a real, let g_i be the Cohen subset of ω_1 obtained at the selected coordinate after applying this isomorphism. Then $\dot{A}_i$ is a name for $\mathbb{A}^n(Y_n(u_i, g_i))$, possibly restricted below a condition named over $P_i^n * C_{E_i}^n$. If there is no coding at stage i, its second factor is trivial.

The proof of Lemma 2.9 applies with M_n in place of L. The forcing C_E^n splits over disjoint index sets in M_n. Given u_i and g_i, the lower part and the indexed family of selected branches determine Y_n in any model containing these data and M_n. The almost disjoint order on the finite pairs is determined by D^n, Y_n, and the condition below which the forcing is restricted. The isomorphisms used to restrict the Cohen forcing change only a bounded part of each coordinate. Composing these isomorphisms with the ones already specified by the iteration leaves the selected g_i unchanged. This gives the

projections to initial segments, the presentations below conditions, and the extension of a tail defined below a Boolean value by using trivial second factors off that value.

For a product of two tails over the same initial segment, use disjoint sets of Cohen indices and translate the names of each factor by the complete embeddings of its initial segments. Either factor can be taken first, since the translated names give the same g_i, u_i, and Y_n in both orders. If a product $R \times Q_j$ is already given and T is a continuation of Q_j below a condition, retain the reindexing of both factors and extend the reindexing on the Q_j side to the new indices for T. Use the image of the same Boolean value to determine where the almost disjoint factors of T are used. The embeddings of the prescribed initial segments are unchanged, including at limit stages by finite support.

Every such iteration of length at most ω_1, including the preparatory branch product, has a dense suborder of size $\aleph_1$, preserves cardinals, and forces CH. Indeed the Cohen product can be taken before the branch forcing. It is countably closed and has size $\aleph_1$ in M_n. In the resulting extension the branch product is c.c.c. by the same countable-closure and finite-product argument as over L. Over W_n the Cohen product adds no countable sequences of ordinals by Easton's lemma. Each almost disjoint factor is Knaster, including below a condition. The finite-support iteration is therefore c.c.c. and has a dense suborder of size at most $\aleph_1$ consisting of finite functions whose values are almost disjoint conditions. The same argument applies to the product presentations and to tails extended off a Boolean value as above. In particular the complete embeddings between initial segments give small two-step extensions of M_n. To see the size bound for the two-step forcing, pass to dense suborders and use mixed names for the conditions in an enumeration of the dense suborder of the quotient.

Every real in an extension by an iteration of length ω_1 belongs to the extension by some P^n_β, $\beta < \omega_1$, as in Lemma 2.10. After forcing with the full Cohen product, take a nice $R^n_{\omega_1}$-name for the real in the dense suborder of finite functions. Its countable ordinal code belongs to W_n by Easton's lemma. Choose β above all the indices in the supports of conditions in the name. The factors before β depend only on the Cohen coordinates before β and their own initial segments. Thus the dense suborder of R^n_β is the same after forcing with either $C^n_{E_{<\beta}}$ or the full Cohen product. Evaluation of the same name gives the real in the P^n_β-extension.

Theorem 4.16. *In a generic extension of W_n by a coding iteration of length at most ω_1,*

$$\Phi_n(u) \quad \Longleftrightarrow \quad \exists i \ (i \text{ is a coding stage and } u_i = u).$$

Proof. At a coding stage the almost disjoint generic real codes $Y_n(u_i, g_i)$ also when the factor is restricted below a condition. The Cohen subset g_i

is obtained from the specified ground-model isomorphism, so Lemma 4.15 applies and gives the same witness in the final extension.

Conversely, suppose r witnesses $\Phi_n(u)$ and no stage codes u. Fix the generic K for the full Cohen product. In $W_n[K]$ take nice names for r, u using finite functions as conditions. Let J be the countable set of stages obtained by closing their supports under the supports of the names used at those stages. These names specify whether to code a real, which real to code, the selected Cohen coordinate and the associated isomorphism, and any condition below which the almost disjoint factor is restricted. The finite-support iteration on J embeds completely into the full iteration. At a successor in J the name for the factor uses only the earlier stages in J, and at limits the embeddings commute with the finite-support projections. The countable ordinal code for these names belongs to W_n. By the c.c.c. of the branch product it has a name supported by a countable $B \subseteq \omega_1$ in M_n.

By Lemma 4.15 there is an unbounded set h_r such that, for every $\gamma \in h_r$ and $k < \omega$, $L[r, m_\infty^n]$ contains a cofinal branch through $S^n_{\tau_u(\gamma,k)}$. The selected Cohen subsets at different coding stages in J disagree on an unbounded set, since the original coordinates do and the ground-model isomorphisms change them only on bounded sets. Hence their associated sets h_i have bounded pairwise intersections. Choose $\gamma \in h_r$ above the bounds on those intersections and above B. There is at most one coding stage $i \in J$ such that $\gamma \in h_i$. If there is one, then $u_i \neq u$, so choose k with $u_i(k) \neq u(k)$. Otherwise choose any $k < \omega$. Then

$$\xi = \tau_u(\gamma, k) \notin B \cup \bigcup_{\substack{i \in J \\ i \text{ is a coding stage}}} \mathrm{Sel}(h_i, u_i).$$

Work over $M_n[K][H \restriction (\omega_1 \setminus \{\xi\})]$. The ordinal code for the names on J belongs to this model. As in Lemma 2.11, interpret the names on J and use trivial factors from the first stage requiring the missing branch at ξ onward. At each earlier stage all data for Y_n are present in the model. The resulting iteration has the same dense suborder of finite functions in this model and in $W_n[K]$. This follows by induction on the stages, using absoluteness of membership and order for the next factor in common generic extensions by the preceding stages. At a limit each finite support is bounded. The modified iteration embeds completely into the iteration on J, and the embedding identifies the restrictions to the Boolean values on which the missing branch is never required. At successors the factors agree below these values. At a limit the value is the meet of the earlier ones. In the given generic extension, no stage on J requires the branch at ξ. The induced generic for the modified iteration thus gives an extension of $M_n[K][H \restriction (\omega_1 \setminus \{\xi\})]$ containing r and u.

The tree S^n_ξ is Suslin in $M_n[K][H \restriction (\omega_1 \setminus \{\xi\})]$, since the ground-model Cohen product can be taken before the branch forcing. The product of the

modified iteration with S^n_ξ is c.c.c. by the Knaster property of the factors and finite support. Thus S^n_ξ is still Suslin in the extension containing r. This extension also contains m^n_∞, hence $L[r, m^n_\infty]$ and a cofinal branch through S^n_ξ, a contradiction. □

4.5 Transfer of the uniformization construction

Fix a recursive universal enumeration of Π^1_{n+3} formulas with two free real variables, and write A^n_m for the corresponding relations. We use the construction in the proof of Theorem 3.1 with W_n, A^n_m, and Φ_n in place of W, A_m, and Φ. We verify the required properties below.

The preceding arguments give the complete embeddings for initial segments and products, the presentations below conditions, and the extension of products preserving specified initial segments. The size bound gives a set of choices of names for choice functions and tails in W_n. The two quotient generics in a product are mutually generic over the common initial extension. The argument by generic intersection therefore applies both to the choice functions and to the restrictions on coding reals absent from the common extension. The definition of allowability uses the forcing relation for A^n_m with respect to the same embeddings.

For persistence, write

$$(x, y) \notin A^n_m \quad \iff \quad \exists z\, \theta_m(z, x, y), \qquad \theta_m \in \Pi^1_{n+2}.$$

If the extension by one factor supplies a witness z, the same z witnesses nonmembership in the product extension by Theorem 4.6. The same applies when a tail forcing nonmembership is appended to one factor in the proof of preservation after further restrictions on coding. If a selected value fails to belong to A^n_m in the final extension, a witness z belongs to the extension by a bounded initial segment, and the same theorem reflects $\theta_m(z, x, y)$ to that extension. All these inclusions are small two-step extensions by the preceding size bound. Thus the arguments for persistence, the least rank and its minimizers, the first initial segment containing a new minimizer, and the preservation of the selected value apply over M_n.

Theorem 4.17. *Let $1 \leq n < \omega$ and suppose that $M^\#_n$ exists. There is a cardinal-preserving set-generic extension of M_n satisfying* CH *and the Π^1_{n+3}-uniformization property.*

Proof. First force over M_n with the branch product producing W_n, then repeat the construction in the proof of Theorem 3.1, using the properties verified above. The resulting coding iteration has length ω_1. In the final extension put

$$U^n_m(x, y) \quad \iff \quad (x, y) \in A^n_m \ \wedge\ \neg\Phi_n(\langle x, y, m\rangle).$$

The construction selects exactly one value in each nonempty section, and this is the unique value in the section for which $\langle x, y, m\rangle$ is never coded. By Theorem 4.16, U^n_m is the graph of the resulting uniformizing function. Lemma 4.15 gives $\Phi_n \in \Sigma^1_{n+3}$, so $U^n_m \in \Pi^1_{n+3}$. Real parameters can be included in the first argument using the fixed recursive pairing function. The cardinal and continuum assertions follow from the preservation argument above. □

5 Further possible applications and open problems

We close with two directions suggested by the construction. First, the method should have an analogue in the generalized Baire-space context at a regular uncountable cardinal κ. In particular, one may ask whether the Π^1_1-uniformization problem for $\kappa^{<\kappa}$ can consistently have a positive solution.

The methods in this paper leave the following question open:

Question 1. *Can the Π^1_n-uniformization property be forced over L for $n > 3$?*

The method introduced here is so far limited to local effects. It would be interesting to force a less local, or even global, behaviour:

Question 2. *Given a pair $n, m \in \omega$ such that $n \neq m$ and $n \neq m + 1$, can one force a universe in which the Π^1_n- and the Π^1_m-uniformization properties both hold?*

Finally, it seems worthwhile to combine the present method with other forcing constructions in order to obtain models with properties that are usually consequences of determinacy assumptions. We record one paradigmatic question:

Question 3. *Assume some sufficiently big large cardinal notion which can exist in L. Is there a model which satisfies the Π^1_3-uniformization property and in which every projective set of reals is Lebesgue measurable?*

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